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Medical instruments with variable bending stiffness profiles

Abstract

A medical instrument can have an elongated shaft, at least three bending stiffness zones arranged along a length of the elongated shaft, and at least two transition zones that have a variable bending stiffness extending over a length thereof. Each bending stiffness zone can have a bending stiffness that extends over a length of the bending stiffness zone. The variable bending stiffness of each transition zone can vary from a first bending stiffness on a first side of the transition zone to a second bending stiffness on a second side of the transition zone.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 16/280,300, filed Feb. 20, 2019, which claims the benefit of U.S. Provisional Application No. 62/649,460, filed Mar. 28, 2018, which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(1) This application relates to medical instruments. More particularly, this application relates to medical instruments with variable bending stiffness profiles.

BACKGROUND

(2) Medical procedures such as endoscopy (e.g., bronchoscopy) may involve accessing and visualizing the inside of a patient's lumen(s) (e.g., airways) for diagnostic and/or therapeutic purposes.

(3) Bronchoscopy is a medical procedure that allows a physician to examine airways in a patient's lungs, such as bronchi and bronchioles. During the procedure, a thin, flexible tubular tool or instrument, known as a bronchoscope, is inserted into the patient's mouth and passed down the patient's throat into his or her lung airways towards a tissue site identified for subsequent diagnosis and/or treatment.

(4) In certain procedures, a robotically enabled medical system may be used to control the insertion and/or manipulation of the instrument. The robotically enabled medical system may include a robotic arm, or other instrument positioning device, having a manipulator assembly used to control the positioning of the instrument during the procedure.

SUMMARY

(5) In a first aspect, a medical instrument is described. The medical instrument includes an elongated shaft extending between a distal end and a proximal end. The elongated shaft includes at least three bending stiffness zones arranged along a length of the elongated shaft, each bending stiffness zone comprising a bending stiffness that extends over a length of the bending stiffness zone, wherein the bending stiffness of each of the bending stiffness zones is greater or less than the bending stiffness of an adjacent bending stiffness zone, and wherein the bending stiffness of each of the bending stiffness zones is substantially uniform along its length. The elongated shaft also includes at least two transition zones, wherein each transition zone comprises a variable bending stiffness extending a length between each pair of the at least three bending stiffness zones, wherein the variable bending stiffness for each transition zone varies from a first bending stiffness on a first side of the transition zone to a second bending stiffness on a second side of the transition zone.

(6) The medical instrument can also include one or more of the following features in any combination: (a) wherein, for each transition zone, the variable bending stiffness varies from the first bending stiffness on the first side of the transition zone to the second bending stiffness on the

second side of the transition zone with a substantially linear slope; (b) wherein the bending stiffness zone closest to the distal end has a bending stiffness that is less than each of the three other bending stiffness zones; (c) wherein the bending stiffness zone closest to the proximal end has a bending stiffness that is greater than each of the three other bending stiffness zones; (d) wherein, from the proximal end to the distal end, the bending stiffness of each of the at least four bending stiffness zones increases; (e) wherein each of the bending stiffness zones has a minimum length of at least 50 mm; (f) wherein each of the transition zones has a minimum length of at least 10 mm; (g) wherein the elongate shaft comprises an inner layer and an outer layer; (h) wherein the outer layer comprises a braided jacket and the inner layer comprises an endoskeleton; (i) wherein the braided jacket comprises one or more pull wires extending therethrough; (j) wherein the bending stiffness of the braided jacket can be modulated by at least one of the following: jacket material durometer, braid geometry, and braid pic count; (k) wherein the endoskeleton comprises a first portion formed of nitinol and a second portion formed of stainless steel; (l) wherein the second portion formed comprises a first section with coils having at least a first pitch and a second section with coils having at least a second pitch; (m) wherein the second portion formed by stainless steel further comprises a third section formed as a rigid hypotube; (n) wherein the medical instrument comprises an endoscope; (o) wherein the elongate shaft has a bending stiffness profile that is omnidirectional; (p) wherein the elongate shaft has a bending stiffness profile that is stepped; and/or (q) wherein each of the transition zones has a bending stiffness in the form of a gradual slope.

(7) In another aspect, a medical system is described. The medical system includes an elongated member having a length extending between a distal end and a proximal end, the elongated member further comprising n bending stiffness zones of different bending stiffnesses, wherein each of the n bending stiffness zones has a bending stiffness that is substantially uniform along its length. The system also includes a sheath having a sheath length extending between a sheath distal end and a sheath proximal end, the sheath comprising an inner channel. The elongated member is moveable within the inner channel of the sheath such that the elongated member and the sheath form a compound structure having at least $n+1$ bending stiffness zones, wherein each of the $n+1$ bending stiffness zones has a bending stiffness that is substantially uniform along its length.

(8) In some embodiments, the system can include one or more of the following features in any combination: (a) wherein the compound structure comprises at least $n+1$ bending stiffness zones when the scope is advanced through the sheath such that the scope distal end is positioned distally of the sheath distal end; (b) wherein the sheath comprises at least n bending stiffness zones of different bending stiffnesses, wherein each of the n bending stiffness zones has a bending stiffness that is substantially uniform across its length, and wherein the scope can be positioned relative to the sheath such that the compound structure comprises at least $2n$ bending stiffness zones; (c) wherein the scope has four zones of bending stiffness and the compound structure has greater than four zones of bending stiffness; (d) wherein the scope and the sheath are of different lengths; (e) wherein the sheath comprises a plurality of zones of bending stiffness; (f) wherein the sheath comprises at least four zones of bending stiffness; and/or (g) a first robotic arm, the scope attached to the first robotic arm, and a second robotic arm, the sheath attached to the second robotic arm, wherein the first robotic arm is configured to advance or retract the scope relative to the sheath, and the second robotic arm is configured to advance or retract the sheath relative to the scope.

(9) In another aspect, a method for navigating a medical instrument within a patient is described. The method includes: inserting the medical instrument into a patient lumen, wherein the medical instrument comprises: a scope comprising a plurality of bending stiffness zones arranged along a length of the scope, wherein each of the bending stiffness zones has a bending stiffness that is substantially uniform, and a sheath comprising a plurality of bending stiffness zones arranged along a length of the sheath, the sheath further comprising an inner channel, wherein the scope is positioned within the inner channel of the sheath; and modulating a bending stiffness profile of the

medical instrument by adjusting the position of at least one of the scope and the sheath relative to the other of the scope and the sheath.

(10) The method can include one or more of the following features in any combination: (a) wherein modulating the bending stiffness profile of the medical instrument comprises advancing the scope or retracting the sheath such that the scope extends distally from the sheath; (b) wherein modulating the bending stiffness profile of the medical instrument comprises advancing the sheath or retracting the scope such that a distal end of the scope is positioned within the inner channel of the sheath; (c) wherein a distal end of the sheath is positioned distally beyond the distal end of the scope; (d) wherein the distal end of the scope is aligned with a distal end of the sheath; (e) wherein the scope comprises at least four bending stiffness zones, each having a substantially constant bending stiffness a long a length of the zone; (f) wherein the scope comprises at least three zones, each positioned between a pair of the at least four bending stiffness zones; (g) wherein the scope comprises at least three bending stiffness zones, each having a substantially constant bending stiffness a long a length of the zone; (h) wherein the scope comprises at least two zones, each positioned between a pair of the at least four bending stiffness zones; (i) wherein the patient lumen comprises the bronchial airways; (j) wherein a first bending stiffness zone has a bending stiffness that is less relative to the other bending stiffness zones; (k) wherein a third zone of bending stiffness extends through an introducer; and/or (l) wherein a fourth zone of bending stiffness does not extend through the patient lumen, nor through the introducer.

(11) In another aspect, a medical instrument is described. The medical instrument includes an elongated shaft extending between a distal end and a proximal end. The elongated shaft includes a first modality for modulating a bending stiffness of the elongated shaft, wherein a modality of the first modality changes at a first point between the distal end and the proximal end, and a second modality for modulating the bending stiffness of the elongated shaft, wherein a modality of the second modality changes at a second point between the distal end and the proximal end, the second point different than the first point. The first point and the second point are positioned such that a bending stiffness profile of the elongated shaft comprises a first bending stiffness zone having a first bending stiffness, a second bending stiffness zone having a second bending stiffness, and a transition zone positioned between the first bending stiffness zone and the second bending stiffness zone, the transition zone comprising a length over which a bending stiffness of the transition zone transitions from the first bending stiffness to the second bending stiffness.

(12) The medical instrument can include one or more of the following features in any combination: (a) wherein the first modality is disposed in a first layer and the second modality is disposed in a second layer; (b) wherein the first modality and second modality are disposed in the same layer; (c) wherein the first modality comprises a material property; (d) wherein the material property comprises hardness; (e) wherein the second modality comprises a mechanical property; (f) wherein the mechanical property comprises at least one of braid geometry and braid pic count; and/or (g) a third modality for modulating bending stiffness.

(13) In another aspect, a non-transitory computer readable medium storing instructions is described. The instructions are configured to cause a processor of a device to at least: insert a medical instrument into a patient lumen. The medical instrument comprises a scope comprising a plurality of bending stiffness zones arranged along a length of the scope, wherein each of the bending stiffness zones has a bending stiffness that is substantially uniform, and a sheath comprising a plurality of bending stiffness zones arranged along a length of the sheath, the sheath further comprising an inner channel, wherein the scope is positioned within the inner channel of the sheath. The instructions are configured to also cause the processor of the device to at least modulate a bending stiffness profile of the medical instrument by adjusting the position of at least one of the scope and the sheath relative to the other of the scope and the sheath.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosed aspects will hereinafter be described in conjunction with the appended drawings, provided to illustrate and not to limit the disclosed aspects, wherein like designations denote like elements.
- (2) FIG. 1 illustrates an embodiment of a cart-based robotic system arranged for diagnostic and/or therapeutic bronchoscopy procedure(s).
- (3) FIG. 2 depicts further aspects of the robotic system of FIG. 1.
- (4) FIG. 3 illustrates an embodiment of the robotic system of FIG. 1 arranged for ureteroscopy.
- (5) FIG. 4 illustrates an embodiment of the robotic system of FIG. 1 arranged for a vascular procedure.
- (6) FIG. 5 illustrates an embodiment of a table-based robotic system arranged for a bronchoscopy procedure.
- (7) FIG. 6 provides an alternative view of the robotic system of FIG. 5.
- (8) FIG. 7 illustrates an example system configured to stow robotic arm(s).
- (9) FIG. 8 illustrates an embodiment of a table-based robotic system configured for a ureteroscopy procedure.
- (10) FIG. 9 illustrates an embodiment of a table-based robotic system configured for a laparoscopic procedure.
- (11) FIG. 10 illustrates an embodiment of the table-based robotic system of FIGS. 5-9 with pitch or tilt adjustment.
- (12) FIG. 11 provides a detailed illustration of the interface between the table and the column of the table-based robotic system of FIGS. 5-10.
- (13) FIG. 12 illustrates an exemplary instrument driver.
- (14) FIG. 13 illustrates an exemplary medical instrument with a paired instrument driver.
- (15) FIG. 14 illustrates an alternative design for an instrument driver and instrument where the axes of the drive units are parallel to the axis of the elongated shaft of the instrument.
- (16) FIG. 15 depicts a block diagram illustrating a localization system that estimates a location of one or more elements of the robotic systems of FIGS. 1-10, such as the location of the instrument of FIGS. 13 and 14, in accordance to an example embodiment.
- (17) FIG. 16A is a side view of an embodiment of a medical instrument configured as an endoscope and having a variable bending stiffness profile.
- (18) FIG. 16B illustrates an example variable bending stiffness profile of the medical instrument of FIG. 16A.
- (19) FIG. 16C illustrates a cross-sectional view of the medical instrument of FIG. 16A, showing an example multi-modality construction thereof.
- (20) FIG. 16D schematically illustrates an example of how individual modalities of the multi-modality construction can be varied to achieve the example variable bending stiffness profile of FIG. 16B for the medical instrument of FIG. 16A.
- (21) FIG. 17 illustrates a side view of one embodiment of an endoskeleton for a medical instrument with a variable bending stiffness profile.
- (22) FIG. 18A is a side view of an embodiment of a medical instrument configured as a sheath and having a variable bending stiffness profile.
- (23) FIG. 18B illustrates an example variable bending stiffness profile of the medical instrument of FIG. 18A.
- (24) FIG. 18C illustrates a cross-sectional view of the medical instrument of FIG. 18A, showing an example multi-modality construction thereof.
- (25) FIG. 18D schematically illustrates an example of how individual modalities of the multi-

modal construction can be varied to achieve the representative variable bending stiffness profile of FIG. 18B for the medical instrument of FIG. 18A.

(26) FIG. 19A illustrates a top view of an embodiment of a compound structure, including a scope and a sheath, with the distal end of the scope aligned with the distal end of the sheath.

(27) FIG. 19B illustrates example bending stiffness profiles for the scope, the sheath, and the compound structure of FIG. 19A.

(28) FIG. 19C illustrates a top view of an embodiment of the compound structure of FIG. 19A with the distal end of the scope extended beyond the distal end of the sheath.

(29) FIG. 19D illustrates example bending stiffness profiles for the scope, the sheath, and the compound structure of FIG. 19C.

(30) FIGS. 20A-20C present experimental data related to a first embodiment of a compound structure including a scope and sheath configured according to the principles described herein.

(31) FIGS. 21A and 21B present experimental data related to a second embodiment of a compound structure including a scope and sheath configured according to the principles described herein.

(32) FIG. 22 illustrates an example of a medical instrument navigating within airways of a patient's lung according to one embodiment. The medical instrument can include a variable bending stiffness profile.

(33) FIG. 23 is a flowchart illustrating an example method for navigating a medical instrument within a patient lumen.

DETAILED DESCRIPTION

1. Overview

(34) Aspects of the present disclosure may be integrated into a robotically enabled medical system capable of performing a variety of medical procedures, including both minimally invasive, such as laparoscopy, and non-invasive, such as endoscopy, procedures. Among endoscopy procedures, the system may be capable of performing bronchoscopy, ureteroscopy, gastroscopy, etc.

(35) In addition to performing the breadth of procedures, the system may provide additional benefits, such as enhanced imaging and guidance to assist the physician. Additionally, the system may provide the physician with the ability to perform the procedure from an ergonomic position without the need for awkward arm motions and positions. Still further, the system may provide the physician with the ability to perform the procedure with improved ease of use such that one or more of the instruments of the system can be controlled by a single user.

(36) Various embodiments will be described below in conjunction with the drawings for purposes of illustration. It should be appreciated that many other implementations of the disclosed concepts are possible, and various advantages can be achieved with the disclosed implementations. Headings are included herein for reference and to aid in locating various sections. These headings are not intended to limit the scope of the concepts described with respect thereto. Such concepts may have applicability throughout the entire specification.

(37) A. Robotic System—Cart.

(38) The robotically enabled medical system may be configured in a variety of ways depending on the particular procedure. FIG. 1 illustrates an embodiment of a cart-based robotically enabled system 10 arranged for a diagnostic and/or therapeutic bronchoscopy procedure. During a bronchoscopy, the system 10 may comprise a cart 11 having one or more robotic arms 12 to deliver a medical instrument, such as a steerable endoscope 13, which may be a procedure-specific bronchoscope for bronchoscopy, to a natural orifice access point (i.e., the mouth of the patient positioned on a table in the present example) to deliver diagnostic and/or therapeutic tools. As shown, the cart 11 may be positioned proximate to the patient's upper torso in order to provide access to the access point. Similarly, the robotic arms 12 may be actuated to position the bronchoscope relative to the access point. The arrangement in FIG. 1 may also be utilized when performing a gastro-intestinal (GI) procedure with a gastroscope, a specialized endoscope for GI procedures. FIG. 2 depicts an example embodiment of the cart in greater detail.

(39) With continued reference to FIG. 1, once the cart **11** is properly positioned, the robotic arms **12** may insert the steerable endoscope **13** into the patient robotically, manually, or a combination thereof. As shown, the steerable endoscope **13** may comprise at least two telescoping parts, such as an inner leader portion and an outer sheath portion, each portion coupled to a separate instrument driver from the set of instrument drivers **28**, each instrument driver coupled to the distal end of an individual robotic arm. This linear arrangement of the instrument drivers **28**, which facilitates coaxially aligning the leader portion with the sheath portion, creates a “virtual rail” **29** that may be repositioned in space by manipulating the one or more robotic arms **12** into different angles and/or positions. The virtual rails described herein are depicted in the Figures using dashed lines, and accordingly the dashed lines do not depict any physical structure of the system. Translation of the instrument drivers **28** along the virtual rail **29** telescopes the inner leader portion relative to the outer sheath portion or advances or retracts the endoscope **13** from the patient. The angle of the virtual rail **29** may be adjusted, translated, and pivoted based on clinical application or physician preference. For example, in bronchoscopy, the angle and position of the virtual rail **29** as shown represents a compromise between providing physician access to the endoscope **13** while minimizing friction that results from bending the endoscope **13** into the patient's mouth.

(40) The endoscope **13** may be directed down the patient's trachea and lungs after insertion using precise commands from the robotic system until reaching the target destination or operative site. In order to enhance navigation through the patient's lung network and/or reach the desired target, the endoscope **13** may be manipulated to telescopically extend the inner leader portion from the outer sheath portion to obtain enhanced articulation and greater bend radius.

(41) For example, the endoscope **13** may be directed to deliver a biopsy needle to a target, such as, for example, a lesion or nodule within the lungs of a patient. The needle may be deployed down a working channel that runs the length of the endoscope to obtain a tissue sample to be analyzed by a pathologist. Depending on the pathology results, additional tools may be deployed down the working channel of the endoscope for additional biopsies. After identifying a nodule to be malignant, the endoscope **13** may endoscopically deliver tools to resect the potentially cancerous tissue. In some instances, diagnostic and therapeutic treatments may need to be delivered in separate procedures. In those circumstances, the endoscope **13** may also be used to deliver a fiducial to “mark” the location of the target nodule as well. In other instances, diagnostic and therapeutic treatments may be delivered during the same procedure.

(42) The system **10** may also include a movable tower **30**, which may be connected via support cables to the cart **11** to provide support for controls, electronics, fluidics, optics, sensors, and/or power to the cart **11**. Placing such functionality in the tower **30** allows for a smaller form factor cart **11** that may be more easily adjusted and/or re-positioned by an operating physician and his/her staff. Additionally, the division of functionality between the cart/table and the support tower **30** reduces operating room clutter and facilitates improving clinical workflow. While the cart **11** may be positioned close to the patient, the tower **30** may be stowed in a remote location to stay out of the way during a procedure.

(43) In support of the robotic systems described above, the tower **30** may include component(s) of a computer-based control system that stores computer program instructions, for example, within a non-transitory computer-readable storage medium such as a persistent magnetic storage drive, solid state drive, etc. The execution of those instructions, whether the execution occurs in the tower **30** or the cart **11**, may control the entire system or sub-system(s) thereof. For example, when executed by a processor of the computer system, the instructions may cause the components of the robotics system to actuate the relevant carriages and arm mounts, actuate the robotics arms, and control the medical instruments. For example, in response to receiving the control signal, the motors in the joints of the robotics arms may position the arms into a certain posture.

(44) The tower **30** may also include a pump, flow meter, valve control, and/or fluid access in order to provide controlled irrigation and aspiration capabilities to system that may be deployed through

the endoscope **13**. These components may also be controlled using the computer system of tower **30**. In some embodiments, irrigation and aspiration capabilities may be delivered directly to the endoscope **13** through separate cable(s).

(45) The tower **30** may include a voltage and surge protector designed to provide filtered and protected electrical power to the cart **11**, thereby avoiding placement of a power transformer and other auxiliary power components in the cart **11**, resulting in a smaller, more moveable cart **11**.

(46) The tower **30** may also include support equipment for the sensors deployed throughout the robotic system **10**. For example, the tower **30** may include opto-electronics equipment for detecting, receiving, and processing data received from the optical sensors or cameras throughout the robotic system **10**. In combination with the control system, such opto-electronics equipment may be used to generate real-time images for display in any number of consoles deployed throughout the system, including in the tower **30**. Similarly, the tower **30** may also include an electronic subsystem for receiving and processing signals received from deployed electromagnetic (EM) sensors. The tower **30** may also be used to house and position an EM field generator for detection by EM sensors in or on the medical instrument.

(47) The tower **30** may also include a console **31** in addition to other consoles available in the rest of the system. e.g., console mounted on top of the cart. The console **31** may include a user interface and a display screen, such as a touchscreen, for the physician operator. Consoles in system **10** are generally designed to provide both robotic controls as well as pre-operative and real-time information of the procedure, such as navigational and localization information of the endoscope **13**. When the console **31** is not the only console available to the physician, it may be used by a second operator, such as a nurse, to monitor the health or vitals of the patient and the operation of system, as well as provide procedure-specific data, such as navigational and localization information.

(48) The tower **30** may be coupled to the cart **11** and endoscope **13** through one or more cables or connections (not shown). In some embodiments, the support functionality from the tower **30** may be provided through a single cable to the cart **11**, simplifying and de-cluttering the operating room. In other embodiments, specific functionality may be coupled in separate cabling and connections. For example, while power may be provided through a single power cable to the cart, the support for controls, optics, fluidics, and/or navigation may be provided through a separate cable.

(49) FIG. 2 provides a detailed illustration of an embodiment of the cart from the cart-based robotically enabled system shown in FIG. 1. The cart **11** generally includes an elongated support structure **14** (often referred to as a “column”), a cart base **15**, and a console **16** at the top of the column **14**. The column **14** may include one or more carriages, such as a carriage **17** (alternatively “arm support”) for supporting the deployment of one or more robotic arms **12** (three shown in FIG. 2). The carriage **17** may include individually configurable arm mounts that rotate along a perpendicular axis to adjust the base of the robotic arms **12** for better positioning relative to the patient. The carriage **17** also includes a carriage interface **19** that allows the carriage **17** to vertically translate along the column **14**.

(50) The carriage interface **19** is connected to the column **14** through slots, such as slot **20**, that are positioned on opposite sides of the column **14** to guide the vertical translation of the carriage **17**. The slot **20** contains a vertical translation interface to position and hold the carriage at various vertical heights relative to the cart base **15**. Vertical translation of the carriage **17** allows the cart **11** to adjust the reach of the robotic arms **12** to meet a variety of table heights, patient sizes, and physician preferences. Similarly, the individually configurable arm mounts on the carriage **17** allow the robotic arm base **21** of robotic arms **12** to be angled in a variety of configurations.

(51) In some embodiments, the slot **20** may be supplemented with slot covers that are flush and parallel to the slot surface to prevent dirt and fluid ingress into the internal chambers of the column **14** and the vertical translation interface as the carriage **17** vertically translates. The slot covers may be deployed through pairs of spring spools positioned near the vertical top and bottom of the slot

20. The covers are coiled within the spools until deployed to extend and retract from their coiled state as the carriage **17** vertically translates up and down. The spring-loading of the spools provides force to retract the cover into a spool when carriage **17** translates towards the spool, while also maintaining a tight seal when the carriage **17** translates away from the spool. The covers may be connected to the carriage **17** using, for example, brackets in the carriage interface **19** to ensure proper extension and retraction of the cover as the carriage **17** translates.

(52) The column **14** may internally comprise mechanisms, such as gears and motors, that are designed to use a vertically aligned lead screw to translate the carriage **17** in a mechanized fashion in response to control signals generated in response to user inputs, e.g., inputs from the console **16**.

(53) The robotic arms **12** may generally comprise robotic arm bases **21** and end effectors **22**, separated by a series of linkages **23** that are connected by a series of joints **24**, each joint comprising an independent actuator, each actuator comprising an independently controllable motor. Each independently controllable joint represents an independent degree of freedom available to the robotic arm. Each of the arms **12** have seven joints, and thus provide seven degrees of freedom. A multitude of joints result in a multitude of degrees of freedom, allowing for “redundant” degrees of freedom. Redundant degrees of freedom allow the robotic arms **12** to position their respective end effectors **22** at a specific position, orientation, and trajectory in space using different linkage positions and joint angles. This allows for the system to position and direct a medical instrument from a desired point in space while allowing the physician to move the arm joints into a clinically advantageous position away from the patient to create greater access, while avoiding arm collisions.

(54) The cart base **15** balances the weight of the column **14**, carriage **17**, and arms **12** over the floor. Accordingly, the cart base **15** houses heavier components, such as electronics, motors, power supply, as well as components that either enable movement and/or immobilize the cart. For example, the cart base **15** includes rollable wheel-shaped casters **25** that allow for the cart to easily move around the room prior to a procedure. After reaching the appropriate position, the casters **25** may be immobilized using wheel locks to hold the cart **11** in place during the procedure.

(55) Positioned at the vertical end of column **14**, the console **16** allows for both a user interface for receiving user input and a display screen (or a dual-purpose device such as, for example, a touchscreen **26**) to provide the physician user with both pre-operative and intra-operative data. Potential pre-operative data on the touchscreen **26** may include pre-operative plans, navigation and mapping data derived from pre-operative computerized tomography (CT) scans, and/or notes from pre-operative patient interviews. Intra-operative data on display may include optical information provided from the tool, sensor and coordinate information from sensors, as well as vital patient statistics, such as respiration, heart rate, and/or pulse. The console **16** may be positioned and tilted to allow a physician to access the console from the side of the column **14** opposite carriage **17**. From this position, the physician may view the console **16**, robotic arms **12**, and patient while operating the console **16** from behind the cart **11**. As shown, the console **16** also includes a handle **27** to assist with maneuvering and stabilizing cart **11**.

(56) FIG. 3 illustrates an embodiment of a robotically enabled system **10** arranged for ureteroscopy. In a ureteroscopic procedure, the cart **11** may be positioned to deliver a ureteroscope **32**, a procedure-specific endoscope designed to traverse a patient's urethra and ureter, to the lower abdominal area of the patient. In a ureteroscopy, it may be desirable for the ureteroscope **32** to be directly aligned with the patient's urethra to reduce friction and forces on the sensitive anatomy in the area. As shown, the cart **11** may be aligned at the foot of the table to allow the robotic arms **12** to position the ureteroscope **32** for direct linear access to the patient's urethra. From the foot of the table, the robotic arms **12** may insert ureteroscope **32** along the virtual rail **33** directly into the patient's lower abdomen through the urethra.

(57) After insertion into the urethra, using similar control techniques as in bronchoscopy, the ureteroscope **32** may be navigated into the bladder, ureters, and/or kidneys for diagnostic and/or

therapeutic applications. For example, the ureteroscope **32** may be directed into the ureter and kidneys to break up kidney stone build up using laser or ultrasonic lithotripsy device deployed down the working channel of the ureteroscope **32**. After lithotripsy is complete, the resulting stone fragments may be removed using baskets deployed down the ureteroscope **32**.

(58) FIG. **4** illustrates an embodiment of a robotically enabled system similarly arranged for a vascular procedure. In a vascular procedure, the system **10** may be configured such the cart **11** may deliver a medical instrument **34**, such as a steerable catheter, to an access point in the femoral artery in the patient's leg. The femoral artery presents both a larger diameter for navigation as well as relatively less circuitous and tortuous path to the patient's heart, which simplifies navigation. As in a ureteroscopic procedure, the cart **11** may be positioned towards the patient's legs and lower abdomen to allow the robotic arms **12** to provide a virtual rail **35** with direct linear access to the femoral artery access point in the patient's thigh/hip region. After insertion into the artery, the medical instrument **34** may be directed and inserted by translating the instrument drivers **28**. Alternatively, the cart may be positioned around the patient's upper abdomen in order to reach alternative vascular access points, such as, for example, the carotid and brachial arteries near the shoulder and wrist.

(59) B. Robotic System—Table.

(60) Embodiments of the robotically enabled medical system may also incorporate the patient's table. Incorporation of the table reduces the amount of capital equipment within the operating room by removing the cart, which allows greater access to the patient. FIG. **5** illustrates an embodiment of such a robotically enabled system arranged for a bronchoscopy procedure. System **36** includes a support structure or column **37** for supporting platform **38** (shown as a “table” or “bed”) over the floor. Much like in the cart-based systems, the end effectors of the robotic arms **39** of the system **36** comprise instrument drivers **42** that are designed to manipulate an elongated medical instrument, such as a bronchoscope **40** in FIG. **5**, through or along a virtual rail **41** formed from the linear alignment of the instrument drivers **42**. In practice, a C-arm for providing fluoroscopic imaging may be positioned over the patient's upper abdominal area by placing the emitter and detector around table **38**.

(61) FIG. **6** provides an alternative view of the system **36** without the patient and medical instrument for discussion purposes. As shown, the column **37** may include one or more carriages **43** shown as ring-shaped in the system **36**, from which the one or more robotic arms **39** may be based. The carriages **43** may translate along a vertical column interface **44** that runs the length of the column **37** to provide different vantage points from which the robotic arms **39** may be positioned to reach the patient. The carriage(s) **43** may rotate around the column **37** using a mechanical motor positioned within the column **37** to allow the robotic arms **39** to have access to multiples sides of the table **38**, such as, for example, both sides of the patient. In embodiments with multiple carriages, the carriages may be individually positioned on the column and may translate and/or rotate independent of the other carriages. While carriages **43** need not surround the column **37** or even be circular, the ring-shape as shown facilitates rotation of the carriages **43** around the column **37** while maintaining structural balance. Rotation and translation of the carriages **43** allows the system to align the medical instruments, such as endoscopes and laparoscopes, into different access points on the patient.

(62) The arms **39** may be mounted on the carriages through a set of arm mounts **45** comprising a series of joints that may individually rotate and/or telescopically extend to provide additional configurability to the robotic arms **39**. Additionally, the arm mounts **45** may be positioned on the carriages **43** such that, when the carriages **43** are appropriately rotated, the arm mounts **45** may be positioned on either the same side of table **38** (as shown in FIG. **6**), on opposite sides of table **38** (as shown in FIG. **9**), or on adjacent sides of the table **38** (not shown).

(63) The column **37** structurally provides support for the table **38**, and a path for vertical translation of the carriages. Internally, the column **37** may be equipped with lead screws for guiding vertical

translation of the carriages, and motors to mechanize the translation of said carriages based on the lead screws. The column **37** may also convey power and control signals to the carriage **43** and robotic arms **39** mounted thereon.

(64) The table base **46** serves a similar function as the cart base **15** in cart **11** shown in FIG. 2, housing heavier components to balance the table/bed **38**, the column **37**, the carriages **43**, and the robotic arms **39**. The table base **46** may also incorporate rigid casters to provide stability during procedures. Deployed from the bottom of the table base **46**, the casters may extend in opposite directions on both sides of the base **46** and retract when the system **36** needs to be moved.

(65) Continuing with FIG. 6, the system **36** may also include a tower (not shown) that divides the functionality of system **36** between table and tower to reduce the form factor and bulk of the table. As in earlier disclosed embodiments, the tower may provide a variety of support functionalities to table, such as processing, computing, and control capabilities, power, fluidics, and/or optical and sensor processing. The tower may also be movable to be positioned away from the patient to improve physician access and de-clutter the operating room. Additionally, placing components in the tower allows for more storage space in the table base for potential stowage of the robotic arms. The tower may also include a console that provides both a user interface for user input, such as keyboard and/or pendant, as well as a display screen (or touchscreen) for pre-operative and intra-operative information, such as real-time imaging, navigation, and tracking information.

(66) In some embodiments, a table base may stow and store the robotic arms when not in use. FIG. 7 illustrates a system **47** that stows robotic arms in an embodiment of the table-based system. In system **47**, carriages **48** may be vertically translated into base **49** to stow robotic arms **50**, arm mounts **51**, and the carriages **48** within the base **49**. Base covers **52** may be translated and retracted open to deploy the carriages **48**, arm mounts **51**, and arms **50** around column **53**, and closed to stow to protect them when not in use. The base covers **52** may be sealed with a membrane **54** along the edges of its opening to prevent dirt and fluid ingress when closed.

(67) FIG. 8 illustrates an embodiment of a robotically enabled table-based system configured for a ureteroscopy procedure. In a ureteroscopy, the table **38** may include a swivel portion **55** for positioning a patient off-angle from the column **37** and table base **46**. The swivel portion **55** may rotate or pivot around a pivot point (e.g., located below the patient's head) in order to position the bottom portion of the swivel portion **55** away from the column **37**. For example, the pivoting of the swivel portion **55** allows a C-arm (not shown) to be positioned over the patient's lower abdomen without competing for space with the column (not shown) below table **38**. By rotating the carriage **35** (not shown) around the column **37**, the robotic arms **39** may directly insert a ureteroscope **56** along a virtual rail **57** into the patient's groin area to reach the urethra. In a ureteroscopy, stirrups **58** may also be fixed to the swivel portion **55** of the table **38** to support the position of the patient's legs during the procedure and allow clear access to the patient's groin area.

(68) In a laparoscopic procedure, through small incision(s) in the patient's abdominal wall, minimally invasive instruments (elongated in shape to accommodate the size of the one or more incisions) may be inserted into the patient's anatomy. After inflation of the patient's abdominal cavity, the instruments, often referred to as laparoscopes, may be directed to perform surgical or medical tasks, such as grasping, cutting, ablating, suturing, etc. FIG. 9 illustrates an embodiment of a robotically enabled table-based system configured for a laparoscopic procedure. As shown in FIG. 9, the carriages **43** of the system **36** may be rotated and vertically adjusted to position pairs of the robotic arms **39** on opposite sides of the table **38**, such that laparoscopes **59** may be positioned using the arm mounts **45** to be passed through minimal incisions on both sides of the patient to reach his/her abdominal cavity.

(69) To accommodate laparoscopic procedures, the robotically enabled table system may also tilt the platform to a desired angle. FIG. 10 illustrates an embodiment of the robotically enabled medical system with pitch or tilt adjustment. As shown in FIG. 10, the system **36** may accommodate tilt of the table **38** to position one portion of the table at a greater distance from the

floor than the other. Additionally, the arm mounts **45** may rotate to match the tilt such that the arms **39** maintain the same planar relationship with table **38**. To accommodate steeper angles, the column **37** may also include telescoping portions **60** that allow vertical extension of column **37** to keep the table **38** from touching the floor or colliding with base **46**.

(70) FIG. **11** provides a detailed illustration of the interface between the table **38** and the column **37**. Pitch rotation mechanism **61** may be configured to alter the pitch angle of the table **38** relative to the column **37** in multiple degrees of freedom. The pitch rotation mechanism **61** may be enabled by the positioning of orthogonal axes **1, 2** at the column-table interface, each axis actuated by a separate motor **3, 4** responsive to an electrical pitch angle command. Rotation along one screw **5** would enable tilt adjustments in one axis **1**, while rotation along the other screw **6** would enable tilt adjustments along the other axis **2**.

(71) For example, pitch adjustments are particularly useful when trying to position the table in a Trendelenburg position, i.e., position the patient's lower abdomen at a higher position from the floor than the patient's lower abdomen, for lower abdominal surgery. The Trendelenburg position causes the patient's internal organs to slide towards his/her upper abdomen through the force of gravity, clearing out the abdominal cavity for minimally invasive tools to enter and perform lower abdominal surgical or medical procedures, such as laparoscopic prostatectomy.

(72) C. Instrument Driver & Interface.

(73) The end effectors of the system's robotic arms comprise (i) an instrument driver (alternatively referred to as "instrument drive mechanism" or "instrument device manipulator") that incorporate electro-mechanical means for actuating the medical instrument and (ii) a removable or detachable medical instrument which may be devoid of any electro-mechanical components, such as motors. This dichotomy may be driven by the need to sterilize medical instruments used in medical procedures, and the inability to adequately sterilize expensive capital equipment due to their intricate mechanical assemblies and sensitive electronics. Accordingly, the medical instruments may be designed to be detached, removed, and interchanged from the instrument driver (and thus the system) for individual sterilization or disposal by the physician or the physician's staff. In contrast, the instrument drivers need not be changed or sterilized, and may be draped for protection.

(74) FIG. **12** illustrates an example instrument driver. Positioned at the distal end of a robotic arm, instrument driver **62** comprises of one or more drive units **63** arranged with parallel axes to provide controlled torque to a medical instrument via drive shafts **64**. Each drive unit **63** comprises an individual drive shaft **64** for interacting with the instrument, a gear head **65** for converting the motor shaft rotation to a desired torque, a motor **66** for generating the drive torque, an encoder **67** to measure the speed of the motor shaft and provide feedback to the control circuitry, and control circuitry **68** for receiving control signals and actuating the drive unit. Each drive unit **63** being independent controlled and motorized, the instrument driver **62** may provide multiple (four as shown in FIG. **12**) independent drive outputs to the medical instrument. In operation, the control circuitry **68** would receive a control signal, transmit a motor signal to the motor **66**, compare the resulting motor speed as measured by the encoder **67** with the desired speed, and modulate the motor signal to generate the desired torque.

(75) For procedures that require a sterile environment, the robotic system may incorporate a drive interface, such as a sterile adapter connected to a sterile drape, that sits between the instrument driver and the medical instrument. The chief purpose of the sterile adapter is to transfer angular motion from the drive shafts of the instrument driver to the drive inputs of the instrument while maintaining physical separation, and thus sterility, between the drive shafts and drive inputs. Accordingly, an example sterile adapter may comprise of a series of rotational inputs and outputs intended to be mated with the drive shafts of the instrument driver and drive inputs on the instrument. Connected to the sterile adapter, the sterile drape, comprised of a thin, flexible material such as transparent or translucent plastic, is designed to cover the capital equipment, such as the instrument driver, robotic arm, and cart (in a cart-based system) or table (in a table-based system).

Use of the drape would allow the capital equipment to be positioned proximate to the patient while still being located in an area not requiring sterilization (i.e., non-sterile field). On the other side of the sterile drape, the medical instrument may interface with the patient in an area requiring sterilization (i.e., sterile field).

(76) D. Medical Instrument.

(77) FIG. 13 illustrates an example medical instrument with a paired instrument driver. Like other instruments designed for use with a robotic system, medical instrument 70 comprises an elongated shaft 71 (or elongate body) and an instrument base 72. The instrument base 72, also referred to as an “instrument handle” due to its intended design for manual interaction by the physician, may generally comprise rotatable drive inputs 73, e.g., receptacles, pulleys or spools, that are designed to be mated with drive outputs 74 that extend through a drive interface on instrument driver 75 at the distal end of robotic arm 76. When physically connected, latched, and/or coupled, the mated drive inputs 73 of instrument base 72 may share axes of rotation with the drive outputs 74 in the instrument driver 75 to allow the transfer of torque from drive outputs 74 to drive inputs 73. In some embodiments, the drive outputs 74 may comprise splines that are designed to mate with receptacles on the drive inputs 73.

(78) The elongated shaft 71 is designed to be delivered through either an anatomical opening or lumen, e.g., as in endoscopy, or a minimally invasive incision, e.g., as in laparoscopy. The elongated shaft 66 may be either flexible (e.g., having properties similar to an endoscope) or rigid (e.g., having properties similar to a laparoscope) or contain a customized combination of both flexible and rigid portions. When designed for laparoscopy, the distal end of a rigid elongated shaft may be connected to an end effector comprising a jointed wrist formed from a clevis with an axis of rotation and a surgical tool or medical instrument, such as, for example, a grasper or scissors, that may be actuated based on force from the tendons as the drive inputs rotate in response to torque received from the drive outputs 74 of the instrument driver 75. When designed for endoscopy, the distal end of a flexible elongated shaft may include a steerable or controllable bending section that may be articulated and bent based on torque received from the drive outputs 74 of the instrument driver 75.

(79) Torque from the instrument driver 75 is transmitted down the elongated shaft 71 using tendons within the shaft 71. These individual tendons, such as pull wires, may be individually anchored to individual drive inputs 73 within the instrument handle 72. From the handle 72, the tendons are directed down one or more pull lumens within the elongated shaft 71 and anchored at the distal portion of the elongated shaft 71. In laparoscopy, these tendons may be coupled to a distally mounted end effector, such as a wrist, grasper, or scissor. Under such an arrangement, torque exerted on drive inputs 73 would transfer tension to the tendon, thereby causing the end effector to actuate in some way. In laparoscopy, the tendon may cause a joint to rotate about an axis, thereby causing the end effector to move in one direction or another. Alternatively, the tendon may be connected to one or more jaws of a grasper at distal end of the elongated shaft 71, where tension from the tendon cause the grasper to close.

(80) In endoscopy, the tendons may be coupled to a bending or articulating section positioned along the elongated shaft 71 (e.g., at the distal end) via adhesive, control ring, or other mechanical fixation. When fixedly attached to the distal end of a bending section, torque exerted on drive inputs 73 would be transmitted down the tendons, causing the softer, bending section (sometimes referred to as the articulable section or region) to bend or articulate. Along the non-bending sections, it may be advantageous to spiral or helix the individual pull lumens that direct the individual tendons along (or inside) the walls of the endoscope shaft to balance the radial forces that result from tension in the pull wires. The angle of the spiraling and/or spacing there between may be altered or engineered for specific purposes, wherein tighter spiraling exhibits lesser shaft compression under load forces, while lower amounts of spiraling results in greater shaft compression under load forces, but also exhibits limits bending. On the other end of the spectrum,

the pull lumens may be directed parallel to the longitudinal axis of the elongated shaft **71** to allow for controlled articulation in the desired bending or articulable sections.

(81) In endoscopy, the elongated shaft **71** houses a number of components to assist with the robotic procedure. The shaft may comprise of a working channel for deploying surgical tools (or medical instruments), irrigation, and/or aspiration to the operative region at the distal end of the shaft **71**. The shaft **71** may also accommodate wires and/or optical fibers to transfer signals to/from an optical assembly at the distal tip, which may include of an optical camera. The shaft **71** may also accommodate optical fibers to carry light from proximally located light sources, such as light emitting diodes, to the distal end of the shaft.

(82) At the distal end of the instrument **70**, the distal tip may also comprise the opening of a working channel for delivering tools for diagnostic and/or therapy, irrigation, and aspiration to an operative site. The distal tip may also include a port for a camera, such as a fiberscope or a digital camera, to capture images of an internal anatomical space. Relatedly, the distal tip may also include ports for light sources for illuminating the anatomical space when using the camera.

(83) In the example of FIG. **13**, the drive shaft axes, and thus the drive input axes, are orthogonal to the axis of the elongated shaft. This arrangement, however, complicates roll capabilities for the elongated shaft **71**. Rolling the elongated shaft **71** along its axis while keeping the drive inputs **73** static results in undesirable tangling of the tendons as they extend off the drive inputs **73** and enter pull lumens within the elongated shaft **71**. The resulting entanglement of such tendons may disrupt any control algorithms intended to predict movement of the flexible elongated shaft during an endoscopic procedure.

(84) FIG. **14** illustrates an alternative design for an instrument driver and instrument where the axes of the drive units are parallel to the axis of the elongated shaft of the instrument. As shown, a circular instrument driver **80** comprises four drive units with their drive outputs **81** aligned in parallel at the end of a robotic arm **82**. The drive units, and their respective drive outputs **81**, are housed in a rotational assembly **83** of the instrument driver **80** that is driven by one of the drive units within the assembly **83**. In response to torque provided by the rotational drive unit, the rotational assembly **83** rotates along a circular bearing that connects the rotational assembly **83** to the non-rotational portion **84** of the instrument driver. Power and controls signals may be communicated from the non-rotational portion **84** of the instrument driver **80** to the rotational assembly **83** through electrical contacts may be maintained through rotation by a brushed slip ring connection (not shown). In other embodiments, the rotational assembly **83** may be responsive to a separate drive unit that is integrated into the non-rotatable portion **84**, and thus not in parallel to the other drive units. The rotational mechanism **83** allows the instrument driver **80** to rotate the drive units, and their respective drive outputs **81**, as a single unit around an instrument driver axis **85**.

(85) Like earlier disclosed embodiments, an instrument **86** may comprise an elongated shaft portion **88** and an instrument base **87** (shown with a transparent external skin for discussion purposes) comprising a plurality of drive inputs **89** (such as receptacles, pulleys, and spools) that are configured to receive the drive outputs **81** in the instrument driver **80**. Unlike prior disclosed embodiments, instrument shaft **88** extends from the center of instrument base **87** with an axis substantially parallel to the axes of the drive inputs **89**, rather than orthogonal as in the design of FIG. **13**.

(86) When coupled to the rotational assembly **83** of the instrument driver **80**, the medical instrument **86**, comprising instrument base **87** and instrument shaft **88**, rotates in combination with the rotational assembly **83** about the instrument driver axis **85**. Since the instrument shaft **88** is positioned at the center of instrument base **87**, the instrument shaft **88** is coaxial with instrument driver axis **85** when attached. Thus, rotation of the rotational assembly **83** causes the instrument shaft **88** to rotate about its own longitudinal axis. Moreover, as the instrument base **87** rotates with the instrument shaft **88**, any tendons connected to the drive inputs **89** in the instrument base **87** are not tangled during rotation. Accordingly, the parallelism of the axes of the drive outputs **81**, drive

inputs **89**, and instrument shaft **88** allows for the shaft rotation without tangling any control tendons.

(87) E. Navigation and Control.

(88) Traditional endoscopy may involve the use of fluoroscopy (e.g., as may be delivered through a C-arm) and other forms of radiation-based imaging modalities to provide endoluminal guidance to an operator physician. In contrast, the robotic systems contemplated by this disclosure can provide for non-radiation-based navigational and localization means to reduce physician exposure to radiation and reduce the amount of equipment within the operating room. As used herein, the term “localization” may refer to determining and/or monitoring the position of objects in a reference coordinate system. Technologies such as pre-operative mapping, computer vision, real-time EM tracking, and robot command data may be used individually or in combination to achieve a radiation-free operating environment. In other cases, where radiation-based imaging modalities are still used, the pre-operative mapping, computer vision, real-time EM tracking, and robot command data may be used individually or in combination to improve upon the information obtained solely through radiation-based imaging modalities.

(89) FIG. **15** is a block diagram illustrating a localization system **90** that estimates a location of one or more elements of the robotic system, such as the location of the instrument, in accordance to an example embodiment. The localization system **90** may be a set of one or more computer devices configured to execute one or more instructions. The computer devices may be embodied by a processor (or processors) and computer-readable memory in one or more components discussed above. By way of example and not limitation, the computer devices may be in the tower **30** shown in FIG. **1**, the cart shown in FIGS. **1-4**, the beds shown in FIGS. **5-10**, etc.

(90) As shown in FIG. **15**, the localization system **90** may include a localization module **95** that processes input data **91-94** to generate location data **96** for the distal tip of a medical instrument. The location data **96** may be data or logic that represents a location and/or orientation of the distal end of the instrument relative to a frame of reference. The frame of reference can be a frame of reference relative to the anatomy of the patient or to a known object, such as an EM field generator (see discussion below for the EM field generator).

(91) The various input data **91-94** are now described in greater detail. Pre-operative mapping may be accomplished through the use of the collection of low dose CT scans. Pre-operative CT scans are reconstructed into three-dimensional images, which are visualized, e.g. as “slices” of a cutaway view of the patient's internal anatomy. When analyzed in the aggregate, image-based models for anatomical cavities, spaces and structures of the patient's anatomy, such as a patient lung network, may be generated. Techniques such as center-line geometry may be determined and approximated from the CT images to develop a three-dimensional volume of the patient's anatomy, referred to as model data **91** (also referred to as “preoperative model data” when generated using only preoperative CT scans). The use of center-line geometry is discussed in U.S. patent application Ser. No. 14/523,760, the contents of which are herein incorporated in its entirety. Network topological models may also be derived from the CT-images, and are particularly appropriate for bronchoscopy.

(92) In some embodiments, the instrument may be equipped with a camera to provide vision data **92**. The localization module **95** may process the vision data to enable one or more vision-based location tracking. For example, the preoperative model data may be used in conjunction with the vision data **92** to enable computer vision-based tracking of the medical instrument (e.g., an endoscope or an instrument advance through a working channel of the endoscope). For example, using the preoperative model data **91**, the robotic system may generate a library of expected endoscopic images from the model based on the expected path of travel of the endoscope, each image linked to a location within the model. Intra-operatively, this library may be referenced by the robotic system in order to compare real-time images captured at the camera (e.g., a camera at a distal end of the endoscope) to those in the image library to assist localization.

(93) Other computer vision-based tracking techniques use feature tracking to determine motion of the camera, and thus the endoscope. Some features of the localization module **95** may identify circular geometries in the preoperative model data **91** that correspond to anatomical lumens and track the change of those geometries to determine which anatomical lumen was selected, as well as the relative rotational and/or translational motion of the camera. Use of a topological map may further enhance vision-based algorithms or techniques.

(94) Optical flow, another computer vision-based technique, may analyze the displacement and translation of image pixels in a video sequence in the vision data **92** to infer camera movement. Examples of optical flow techniques may include motion detection, object segmentation calculations, luminance, motion compensated encoding, stereo disparity measurement, etc. Through the comparison of multiple frames over multiple iterations, movement and location of the camera (and thus the endoscope) may be determined.

(95) The localization module **95** may use real-time EM tracking to generate a real-time location of the endoscope in a global coordinate system that may be registered to the patient's anatomy, represented by the preoperative model. In EM tracking, an EM sensor (or tracker) comprising of one or more sensor coils embedded in one or more locations and orientations in a medical instrument (e.g., an endoscopic tool) measures the variation in the EM field created by one or more static EM field generators positioned at a known location. The location information detected by the EM sensors is stored as EM data **93**. The EM field generator (or transmitter), may be placed close to the patient to create a low intensity magnetic field that the embedded sensor may detect. The magnetic field induces small currents in the sensor coils of the EM sensor, which may be analyzed to determine the distance and angle between the EM sensor and the EM field generator. These distances and orientations may be intra-operatively "registered" to the patient anatomy (e.g., the preoperative model) in order to determine the geometric transformation that aligns a single location in the coordinate system with a position in the pre-operative model of the patient's anatomy. Once registered, an embedded EM tracker in one or more positions of the medical instrument (e.g., the distal tip of an endoscope) may provide real-time indications of the progression of the medical instrument through the patient's anatomy.

(96) Robotic command and kinematics data **94** may also be used by the localization module **95** to provide localization data **96** for the robotic system. Device pitch and yaw resulting from articulation commands may be determined during pre-operative calibration. Intra-operatively, these calibration measurements may be used in combination with known insertion depth information to estimate the position of the instrument. Alternatively, these calculations may be analyzed in combination with EM, vision, and/or topological modeling to estimate the position of the medical instrument within the network.

(97) As FIG. **15** shows, a number of other input data can be used by the localization module **95**. For example, although not shown in FIG. **15**, an instrument utilizing shape-sensing fiber can provide shape data that the localization module **95** can use to determine the location and shape of the instrument.

(98) The localization module **95** may use the input data **91-94** in combination(s). In some cases, such a combination may use a probabilistic approach where the localization module **95** assigns a confidence weight to the location determined from each of the input data **91-94**. Thus, where the EM data may not be reliable (as may be the case where there is EM interference) the confidence of the location determined by the EM data **93** can be decrease and the localization module **95** may rely more heavily on the vision data **92** and/or the robotic command and kinematics data **94**.

(99) As discussed above, the robotic systems discussed herein may be designed to incorporate a combination of one or more of the technologies above. The robotic system's computer-based control system, based in the tower, bed and/or cart, may store computer program instructions, for example, within a non-transitory computer-readable storage medium such as a persistent magnetic storage drive, solid state drive, or the like, that, upon execution, cause the system to receive and

analyze sensor data and user commands, generate control signals throughout the system, and display the navigational and localization data, such as the position of the instrument within the global coordinate system, anatomical map, etc.

2. Medical Instruments with Variable Bending Stiffness Profiles

(100) This section relates to medical instruments with variable bending stiffness profiles (also referred to in this application as “variable bending stiffness medical instruments” or in some instances simply as “medical instruments”). In some embodiments, the medical instruments with variable bending stiffness profiles can be used with the robotically enabled medical systems described above with reference to FIGS. 1-15. For example, in some embodiments, any of the medical instruments described above (e.g., endoscope **13**, ureteroscope **32**, medical instrument **34**, bronchoscope **40**, ureteroscope **56**, medical instrument **70**, and others) can include a variable bending stiffness profile as described herein. In addition to robotic implementations, the medical instruments with variable bending stiffness profiles can also be configured for manual use (i.e., non-robotic use).

(101) The medical instruments with variable bending stiffness profiles can be useful for navigating through tortuous paths within a patient's anatomy. In some embodiments, the medical instruments with variable bending stiffness profiles can be particularly useful in navigating through the pulmonary airways of a patient. The pulmonary airways can be tortuous paths. During some medical procedures the medical instruments can be navigated through the pulmonary airways to detect, diagnose, and/or treat an abnormal growth, such as a tumor. Scopes that are sent into the pulmonary airways often want to travel down incorrect pathways and may struggle to enter into the correct pathways. The medical instruments described herein can be advantageously capable of navigating through the correct pathways due to their variable bending stiffness profiles.

(102) FIG. 22 illustrates an example of a medical instrument **100** navigating within a patient's lung. The instrument can be, for example, a scope, a sheath, or a compound instrument comprising a scope positioned within an inner channel of the sheath. In the illustrated example, the instrument **100** is navigated through airways in the lung toward a target site **182** in an upper lobe **178** of the lung. As shown, the airways comprise tortuous paths. As the instrument **100** is inserted into the lung, a distal end of the instrument may be articulated to guide the instrument into particular pathways. However, such navigation can be difficult due to the tortuosity of the lung.

(103) The present application describes medical instruments with variable bending stiffness that are capable of extending through tortuous paths in a patient, including but not limited to the pulmonary airways. In some embodiments, a medical instrument with variable bending stiffness can comprises an elongated shaft. In other embodiments, a medical instrument with variable bending stiffness can be an elongated shaft positioned within an inner channel of a catheter or sheath. The elongated shaft can comprise a leader or scope.

(104) In some embodiments, a variable bending stiffness medical instrument can include an elongated shaft configured for insertion into a patient during a medical procedure that includes a plurality of sections having different bending stiffness properties. As an initial example, a variable bending stiffness medical instrument can include a distal section that has a lower bending stiffness (e.g., that flexes or bends more easily and/or requires less force to flex or bend) than a bending stiffness of a proximal section. In this example, the distal section is more easily bendable so as to facilitate steering or navigation of the medical instrument, while the proximal section is stiff to facilitate pushing the medical instrument through a patient lumen.

(105) Continuing this example, the distal section may include a bending stiffness zone having a substantially constant bending stiffness along its length. The proximal section may also include a bending stiffness zone having a substantially constant bending stiffness along its length. The substantially constant bending stiffness of the distal section can be lower than the bending stiffness of the proximal section (although, in some embodiments, the reverse may also be true).

(106) The variable bending stiffness medical instrument of this initial example can also include a

transition zone between the bending stiffness zone of the distal section and the bending stiffness zone of the proximal section. The transition zone may comprise a length over which the bending stiffness gradually varies, for example, from the substantially constant bending stiffness of the bending stiffness zone of the distal section to the substantially constant bending stiffness of the bending stiffness zone of proximal section. In some embodiments, the bending stiffness of the transition section varies at a generally linear slope, rate, or gradient, although this need not be the case in all embodiments. In some embodiments, the slope of the transition zone is substantially constant.

(107) Thus, as shown by the initial example, a variable bending stiffness instrument can include a plurality of bending stiffness zones, each having a bending stiffness that is substantially constant along a length of the bending stiffness zone, and one or more transition zones, each positioned between an adjacent pair of bending stiffness zones in which the bending stiffness gradually transitions between the pair of bending stiffness zones.

(108) Plotting the bending stiffness along the length of the elongated shaft of the variable bending stiffness medical instrument produces a bending stiffness profile. Graphically, each bending stiffness zone (having a substantially constant bending stiffness along its length) is shown as a plateau within the bending stiffness profile, and each transition zone is shown as a slope or ramp (either generally increasing or generally decreasing) between adjacent plateaus. See, for example, FIG. 16B, described in detail below.

(109) The materials and methods of construction (often referred to herein as modalities or properties) of the elongated shaft of a medical instrument determine the bending stiffness at each point along its length. For example, manufacturing the elongated shaft from a stiffer material will produce a higher bending stiffness.

(110) In some embodiments, to produce a medical instrument with a variable bending stiffness, the materials and/or methods of construction can be varied along the length of the elongated shaft. For example, in the embodiment described above, a stiffer material can be used in the proximal section and a more flexible material can be used in the distal section. However, merely varying a single modality (e.g., material or method of construction) along the length may not produce a bending profile having bending stiffness zones and transitions zones as described above. For example, if the only variation in the elongated shaft is the material change described above (a stiff proximal section and a more flexible distal section) the bending stiffness profile would likely comprise only two plateaus connected by a vertical jump. It may not comprise a transition zone having a length over which the bending stiffness gradually transitions. This may disadvantageously create a stress riser or failure point at the transition between the materials.

(111) Thus, as described more fully below, a variable bending stiffness medical instrument may include a construction that varies, in a staggered manner, in more than one modality of the elongated shaft in each transition zone. This may distribute stresses associated with modality changes along a length of the elongated shaft and produce a more gradual (e.g., sloped or ramped) transition zone as described above.

(112) In some embodiments, a medical instrument with a variable bending stiffness can be configured as a scope (also referred to as a leader), such as an endoscope. The scope can be configured to be inserted through a working or inner channel of a sheath, for example, as shown in FIGS. 19A and 19C. That is, the scope can telescope within the inner channel of the sheath. In some embodiments, the sheath can also be configured as a medical instrument with a variable bending stiffness. Together, the scope and sheath can form a compound structure. In some embodiments, the compound structure has a bending stiffness profile that is different from the bending stiffness profile of either the scope or sheath alone. With the scope positioned within the inner channel of the sheath, the compound structure may comprise a bending profile that is a combination of the individual bending profiles of the scope and the sheath. One skilled in the art will appreciate that the term “compound structure” encompasses all parts of the scope and the

sheath whenever the scope is positioned within the inner channel of the sheath, including parts of the scope that may extend distally from the sheath, or parts of the sheath that may extend distally from the scope.

(113) Further, as will be described more fully below, it is possible to modulate (i.e., change or vary) the bending stiffness profile of the compound structure by moving the scope relative to the sheath, or vice versa. For example, both the scope and the sheath can include a plurality of different bending stiffness zones separated by transition zones. When, for example, the distal tip of the scope is aligned with the distal tip of the sheath, the plurality of different bending stiffness zones and transition zones line up in a first way to produce a first combined bending stiffness profile for the compound structure. When, for example, the scope is further inserted through the sheath such that the distal tip of the scope extends beyond the distal tip of the sheath, the plurality of different bending stiffness zones and transition zones line up in a second way to produce a second combined bending stiffness profile that may be different than the first combined bending stiffness profile.

(114) In some embodiments, the first combined bending stiffness profile may include a first number of bending stiffness zones and the second combined bending stiffness profile may include a second number of bending stiffness zones. Thus, it is possible to vary the number of bending stiffness zones of the compound structure by moving the scope relative to the sheath or vice versa. Although two bending stiffness profiles for the compound structure are mentioned in this example, it will be appreciated that there are many ways in which the scope can be positioned relative to the sheath (e.g., by varying the relative distance between the distal end of the scope and the distal end of the sheath) and each of these can produce a unique bending stiffness profile for the compound structure.

(115) In some embodiments, a compound structure may comprise a bending stiffness profile wherein the bending stiffness of the compound structure varies depending on the position of the sheath relative to the scope. As the bending stiffness varies, the bending stiffness may assume a range or area of coverage along the length of the scope. Different regions of bending stiffness are shown below, such as for example in FIG. 16D.

(116) In some embodiments that include a compound structure having a scope positioned within an inner channel of a sheath, the scope and the sheath may comprise different lengths. For example, the scope and the sheath might not be coextensive in length. In some embodiments, the sheath is shorter than the scope.

(117) The ability to modulate or change the bending stiffness profile of the compound structure (e.g., the combined scope and sheath) can be advantageous for various reasons. For example, during a procedure, a physician (or in some instances, a robotically enabled medical system) can adjust the bending stiffness profile of the compound structure in order to facilitate navigation of a particular portion of a patient's anatomy. This can be accomplished by, for example, moving the scope relative to the sheath.

(118) As a more particular example, during bronchoscopy, a physician may examine airways in a patient's lungs, such as bronchi and bronchioles. For example, as shown in FIG. 22, during the procedure, a medical instrument **100** (e.g., a scope and sheath) may be inserted into the patient's mouth and passed down the patient's trachea into his or her lung airways towards a tissue site identified for subsequent diagnosis and/or treatment (e.g., target **182**). The pulmonary airways are tortuous paths that can be particularly difficult to navigate. It can be particularly challenging to navigate the secondary bronchi, tertiary bronchi, bronchioles, and the upper lobes **178** of the lung. Often, navigating in these portions of the lung requires the medical instrument to make tight turns. The physician may find that a particular bending stiffness profile of the instrument is not well suited for a particular turn or maneuver. For example, with the particular bending stiffness profile the medical instrument may be inclined to travel down an incorrect pathway and/or may struggle to enter a correct pathway. In such a situation, the physician can modulate or change the bending stiffness profile of the medical instrument by moving the scope relative to the sheath (e.g., inserting

the scope further through the sheath such that the scope extends from the sheath, positioning the scope and sheath such that their distal ends are aligned, or retracting the scope within the sheath such that the scope extends beyond the sheath) to modulate the bending stiffness profile. The physician can position the scope and the sheath such that the medical instrument has a bending stiffness profile better suited for the desired maneuver and then perform the maneuver.

(119) These and other features and advantages of medical instruments with variable bending stiffnesses will now be described in greater detail with reference to several embodiments illustrated in the figures. The illustrated embodiments are provided by way of example and are not intended to be limiting.

A. Example Medical Instruments with Variable Bending Stiffness Profiles

(120) FIGS. **16A-16D** relate to an embodiment of a medical instrument **100** having an example variable bending stiffness profile **102**. In the illustrated embodiment, the medical instrument **100** is configured as a catheter, such as a scope (e.g., an endoscope or bronchoscope). The principles and features described with reference to this illustrated example, however, may be applicable to other types of medical instruments as well, for example, other types of scopes (e.g., ureteroscopes, bronchoscopes, etc.), sheaths, etc. FIG. **16A** is a side view of the medical instrument **100**. FIG. **16B** illustrates an example variable bending stiffness profile **102** of the medical instrument **100**. FIG. **16C** illustrates a cross-sectional view of the medical instrument **100** showing an example multi-modality construction thereof. FIG. **16D** schematically illustrates an example of how the individual modalities of the multi-modality construction can be varied to achieve the variable bending stiffness profile **102** of the medical instrument **100**.

(121) As illustrated in FIG. **16A**, the medical instrument **100** includes an instrument base **104** and an elongated shaft **106**. The instrument base **104**, which can also be referred to as a handle, can be configured to attach to an instrument drive mechanism, for example, as shown in FIG. **13** (showing an example instrument base **72** configured to attach to an instrument drive mechanism **75**). The instrument drive mechanism may electronically and/or mechanically couple to the instrument base **104** such that the medical instrument **100** can be robotically controlled. Although the illustrated embodiment is configured for robotic control, in some embodiments, the instrument base **104** may be replaced with an instrument handle configured to enable manual operation and control of the medical instrument **100**.

(122) The elongated shaft **106** can be configured for insertion into a patient during a medical procedure. For example, the elongated shaft **106** can be designed to be delivered through either an anatomical opening or lumen, for example, as in endoscopy, or a minimally invasive incision, for example, as in laparoscopy. In the case of bronchoscopy, the elongated shaft **106** can be inserted through an introducer and into a patient's mouth, down the trachea, and into the lung. To navigate within the patient's body, at least a portion of the elongated shaft **106** may be flexible. In some embodiments, the elongated shaft **106** can include one or more pull wires **108** (for example, as shown in FIG. **16C**). The one or more pull wires **108** can be actuatable to bend one or more portions of the elongated shaft **106** in order to control the shape or pose thereof. For example, during a procedure, the one or more pull wires **108** can be actuated to guide the elongated shaft **106** through the patient's anatomy to a target site. In the case of bronchoscopy, this can involve guiding the elongated shaft through a complex network of branching airways. In some embodiments, the elongated shaft **106** can comprise one or more sensors to assist in localization and/or navigation of the elongated shaft **106** in a patient (for example, as described above with reference to FIG. **15**). In addition or alternatively, in some embodiments, the elongated shaft **106** can include one or more optical fibers for determining a shape of the elongated shaft **106**.

(123) The elongated shaft **106** can extend between a proximal end **110** and a distal end **112**. The proximal end **110** can extend from and be attached to the instrument base **104**. The distal end **112** can be the leading end of the elongated shaft **106** of the medical instrument **100**. For example, the distal end **112** can be the first point introduced into the patient, and, during a medical procedure, the

medical instrument **100** can be guided or driven such that the distal end **112** is advanced toward a target site within the patient. In some embodiments, the distal end **112** includes various features for visualizing and/or treating the target site. For example, the distal end **112** can include a camera, a light source, an end effector (e.g., graspers, cutters, basketing device, etc.), and/or an opening to a working channel through which additional medical tools or instruments can be advanced. Medical tools or instruments that can be delivered through the working channel include, but are not limited to, guidewires, diagnostic and biopsy tools (e.g., ultrasound devices, wires, brushes and other dithering and non-dithering tools), and tools for delivering therapeutic effects (e.g., tools for brachytherapy). While in some embodiments, the camera can be delivered down a channel that is different from other tools or instruments, in other embodiments, the camera can be delivered through the same working channel as other medical tools or instruments. For example, in some embodiments, the camera can be delivered through a working channel in the scope and subsequently removed to make room for a diagnostic and/or treatment tools (such as a dithering or non-dithering tool) to detect and/or treat an abnormal growth. Certain features are illustrated in FIG. **16C**, described in greater detail below.

(124) The elongated shaft **106** includes a length *L* measured between the proximal end **110** and the distal end **112**. The length *L* may be selected so as to facilitate a medical procedure or a range of medical procedures for which the medical instrument **100** will be used. Relevant factors for selecting the length *L* may include path length to the target site, patient anatomy, patient size, drivability and control, operating environment conditions (e.g., placement or robotic arms within an operating room), etc. In some embodiments, for example, the length *L* may be between about 700 and 1200 mm, between about 800 and 1100 mm, or between about 900 and 1000 mm. In one example, the length *L* is about 930 mm.

(125) In the illustrated embodiment, the elongated shaft **106** includes four different sections, a bending section (also referred to an active bending section) **114**, a passive bending section **116**, an internal shaft section **118** (also referred to as a tracking section), and an external shaft section **120**. The sections can be arranged serially along the length *L* of the elongated shaft **106** between the distal end **112** and the proximal end **110**. Although four sections are illustrated, in some embodiments, a medical instrument can include more or less sections. For example, a medical instrument can include one, two, three, four, five, six, seven, or more sections.

(126) Each section can be configured to provide a different flexibility or bending stiffness to facilitate the medical procedure and/or improve the drivability and control of the medical instrument **100**. For example, in some embodiments, more distally located sections (e.g., active bending section **114** and passive bending section **116**) are more flexible to facilitate maneuverability of the elongated shaft **106**, and more proximally located sections (e.g., internal shaft section **118** and external shaft section **120**) are stiffer to provide pushability for the elongated shaft **106**, while preventing or reducing the likelihood of buckling. Further, in some embodiments, the active bending section **114** can be more flexible than the passive bending section **116**, and the external shaft section **120** can be stiffer than the internal shaft section **118**. Other arrangements are also possible.

(127) In some embodiments, the active bending section **114** provides steerability to the distal end of the medical instrument **100**. In some embodiments, the passive bending section **116** provides enough flexibility to advantageously track up into the peripheral or upper lobes of the lung, for example, in the case of the bronchoscopy. In some embodiments, the internal shaft section **118** is flexible enough to be inserted through an introducer and tube bends, but pushable/rigid enough to provide support for the elongated shaft **106**. In some embodiments, the external shaft section **120** is a part of the elongated shaft **106** that remains external to the patient during a procedure (e.g., does not enter the introducer) and is stiff enough to provide support for the distal end **112** of the elongated shaft **106** and prevent buckling.

(128) In some embodiments, the active bending section **114** can be between about 50 and 80 mm,

or approximately 65 mm. In some embodiments, the active bending section **114** defines the articulation radius of the scope. In some embodiments, the passive bending section **116** can be between about 150 mm and 190 mm, or approximately 170 mm. In some embodiments, the internal shaft section **118** can be between about 540 mm and 600 mm, or approximately 572 mm. In some embodiments, the external shaft section **120** can be between about 90 mm and 150 mm, or approximately 123 mm. Advantageously, the different lengths of the four different zones are designed to desirably align the elongated shaft **106** with portions of an overlying sheath, thereby creating a compound structure with desired variable stiffnesses along different lengths as described below. In some embodiments, other lengths for one, more, or all of these sections are also possible. (129) In some embodiments, to achieve these different functions, each section comprises different bending stiffness characteristics or properties. The preceding paragraphs have described the sections in terms of flexibility and stiffness. It will be appreciated that each of these properties is related to bending stiffness. For example, bending stiffness is generally inversely related to flexibility: as bending stiffness increases, flexibility decreases. Bending stiffness is also generally proportionally related to stiffness: as bending stiffness increases, stiffness increases.

(130) FIG. **16B** illustrates a representative bending stiffness profile **102** of the elongated shaft **106** of the medical instrument **100**. The bending stiffness profile **102** illustrates the bending stiffness of the elongated shaft **106** as a function of or along its length *L* between the distal end **112** and the proximal end **110**. In the illustrated example, the x-axis represents the length *L* of the elongated shaft **106** between the distal end **112** and the proximal end **110**, and the y-axis is a measure of the max-force bending stiffness of the elongated shaft **106** measured in Newtons (N). In some examples, to calculate the max-force bending stiffness, a cantilever bend test is performed. In a cantilever bend test, a force tester (e.g., an anvil) is used to push down (bend) the elongated shaft **106** to a fixed distance. The force required to bend the elongated shaft **106** to the fixed distance is measured as the max-force bending stiffness, although technically, the force that is measured is the reaction force from the elongated shaft **106**. In general, the lower the max-force bending stiffness, the more flexible the scope.

(131) As shown by the example bending stiffness profile **102** of FIG. **16B**, the bending stiffness of the elongated shaft **106** varies or changes along its length. Accordingly, the medical instrument **100** comprises a variable bending stiffness profile **102**.

(132) In the illustrated embodiment and as seen in the example bending stiffness profile **102**, the elongated shaft **106** of the medical instrument **100** comprises a plurality of bending stiffness zones wherein the bending stiffness is substantially constant (represented as plateaus within the bending stiffness profile **102**) and a plurality of transition zones (represented as ramps within the bending stiffness profile **102**). As illustrated, the medical instrument **100** includes four bending stiffness zones of substantially constant bending stiffness and three transition zones, with each transition zone positioned between a pair of adjacent bending stiffness zones. In the illustrated example, the medical instrument **100** includes (arranged distally to proximally) a first bending stiffness zone **122** of substantially constant bending stiffness, a first transition zone **124**, a second bending stiffness zone **126** of substantially constant bending stiffness, a second transition zone **128**, a third bending stiffness zone **130** of substantially constant bending stiffness, a third transition zone **132**, and a fourth bending stiffness zone **134** of substantially constant bending stiffness.

(133) In some embodiments, each section can be viewed as a zone of bending stiffness, such that the elongated shaft **106** includes four zones of bending stiffness, with each zone from the most distal bending section **114** to the most proximal section **120** increasing in stiffness. In some embodiments, the first bending stiffness zone **122** has a bending stiffness between about 1 N to 5 N; the second bending stiffness zone **126** has a bending stiffness between about 5 N to 10 N; the third bending stiffness zone **130** has a bending stiffness between about 15 N to 25 N; and the fourth bending stiffness zone **134** has a bending stiffness greater than 30 N. In some embodiments, one, more, or all of the bending stiffness zones can comprise other bending stiffnesses (e.g., higher or

lower) than the listed values.

(134) Although four bending stiffness zones and three transition zones are shown and described, other embodiments may include other numbers of bending stiffness zones and transition zones. For example, a medical instrument could include two, three, four, five, six, or more bending stiffness zones of substantially constant bending stiffness, separated by one, two, three, four, five, or more transition zones.

(135) As used herein, the term “bending stiffness zone having a substantially constant bending stiffness” (sometimes referred to only as a “bending stiffness zone”) refers to a length of the elongated shaft **106** over which the bending stiffness remains substantially constant. In some examples, the length may be between approximately 10 mm and 200 mm. In some embodiments, the length is approximately 10 mm, 25 mm, 50 mm, 75 mm, 100 mm, 150 mm, 200 mm, 300 mm, 400 mm, 500 mm or longer. In some examples, “substantially constant” means that, over the length, the bending stiffness varies by less than about 2.5%, 5%, or 10%, the average bending stiffness value of the elongated shaft **106** over the length. In some examples, “substantially constant” means that, over the length, the bending stiffness varies by less than a threshold force value, such as 0.1 N, 0.25 N, 0.5 N, 1 N, 5 N. or 10 N.

(136) As used herein, “transition zone” refers to a length of the elongated shaft **106** over which the bending stiffness varies or transitions from one value to another value. In some examples, the length may be between approximately 10 mm and 200 mm. In some embodiments, the length is between 10 mm and 200 mm, or approximately 10 mm, 25 mm, 50 mm, 75 mm, 100 mm, 150 mm, 200 mm, 300 mm, 400 mm, 500 mm or longer. In some embodiments, a transition zone is positioned between a pair of bending stiffness zones of substantially constant bending stiffness, and the transition zone comprises a length over which the bending stiffness transitions from the substantially constant bending stiffness of the bending stiffness zone on one side of the transition zone to the substantially constant bending stiffness of the bending stiffness zone of the bending stiffness zone on the other side of the transition zone.

(137) In some embodiments, within a transition zone, the bending stiffness may vary at a generally or substantially constant or linear rate such that the slope of the transition zone is substantially constant. In this context, in some examples, “substantially constant” means that, over the length of the transition zone, the slope varies by less than about 2.5%, 5%, or 10%, the average slope value of the elongated shaft **106** over the length of the transition zone. In some embodiments, it may be preferred to have transition zones wherein the bending stiffness varies at a generally or substantially constant or linear rate. However, transition zones need not be linear in all embodiments. In some embodiments, the transition zones have a curved ramp shape.

(138) In the illustrated embodiment, the first bending stiffness zone **122** has the lowest substantially constant bending stiffness, the second bending stiffness zone **126** has a substantially constant bending stiffness that is higher than that of the first bending stiffness zone **122**, the third bending stiffness zone **130** has a substantially constant bending stiffness that is higher than that of the second bending stiffness zone **126**, and the fourth bending stiffness zone **134** has a substantially constant bending stiffness that is higher than that of the third bending stiffness zone **130**. Thus, the bending stiffness of the medical instrument **100** can generally increase from the distal end **112** to the proximal end **110**. This need not be the case in all embodiments. For example, the bending stiffness could increase, decrease, and then increase again, or generally decrease.

(139) Further, in the illustrated embodiment, the transition zones **124**, **128**, **132** include generally or substantially constant slopes. It will be appreciated that the slope of each transition zone can vary. For example, the slope of the first transition zone **124** can be less than the slope of the second transition zone **128**, and the slope of the second transition zone **128** can be less than the slope of the third transition zone **132**. Again, this need not be the case in all embodiments.

(140) For reference, FIG. **16B** also illustrates how, in one example, the bending stiffness profile **102** aligns with the active bending section **114**, the passive bending section **116**, the internal shaft

section **118**, and the external shaft section **120**. In the illustrated embodiment, each section includes one bending stiffness zone of substantially constant bending stiffness. For example, the first bending stiffness zone **122** is positioned in the active bending section **114**, the second bending stiffness zone **126** is positioned in the passive bending section **116**, the third bending stiffness zone **130** is positioned in the internal shaft section **118**, and fourth bending stiffness zone **134** is positioned in the external shaft section **120**. The elongated shaft **106** can be configured such that the substantially constant bending stiffness of the bending stiffness zone positioned within the section facilitates or enables the section to perform its specified function.

(141) The transition zones **124**, **128**, **132** provide lengths over which the bending stiffness of the elongated shaft **106** can gradually vary from a lower value to a higher value (or vice versa). Having regions where the bending stiffness varies gradually (as opposed to abruptly) can help to avoid stress risers and failure points within the elongated shaft **106**.

(142) In some embodiments, each section includes at least one bending stiffness zone of substantially constant bending stiffness. In some embodiments, each section includes exactly one bending stiffness zone of substantially constant bending stiffness. In some embodiments, one or more sections include more than one bending stiffness zone of substantially constant bending stiffness.

(143) In the illustrated embodiment, the transition zones **124**, **128**, **132** are positioned to overlap adjacent sections. For example, the first transition zone **124** is positioned between and partially overlaps the active bending section **114** and the passive bending section **116**. The second transition zone **128** is positioned between and partially overlaps the passive bending section **116** and the internal shaft section **118**. And the third transition zone **132** is positioned between and partially overlaps the internal shaft section **118** and the external shaft section **120**. Other arrangements and positions for the transition zones are also possible. For example, a transition zone could be positioned entirely within one section, for example, at a distal end of the section, at a proximal end of the section, or along the length of a section between the proximal and distal ends.

(144) As mentioned above, the materials and methods of construction (referred to herein as modalities or properties) of the elongated shaft **106** of a medical instrument **10** determine the bending stiffness, and to produce a medical instrument **100** with a variable bending stiffness (e.g., with a bending stiffness profile as illustrated in FIG. **16**), the materials and/or methods of construction can be varied along the length of the elongated shaft **106**. Further, to achieve the gradual transition zones described above, the elongated shaft **106** can be manufactured with a plurality of modalities and that are varied in a staggered manner. Example modalities and arrangements of modalities to produce variable bending stiffness profiles will now be described with reference to FIGS. **16C** and **16D**.

(145) FIG. **16C** is a cross-sectional view of the elongated shaft **106** of medical instrument **100** including a multi-modality construction configured to provide a variable bending stiffness profile **102** as shown, for example, in FIG. **16C**. The multi-modality construction of the medical instrument **100** can comprise a plurality of layers. For example, as illustrated, the medical instrument **100** comprises an inner layer **144** and an outer layer **146**. The inner and outer layers **144**, **146** can surround various other features that enable some of the functionality of the medical instrument **100**. For example, as illustrated in FIGS. **16A-16D**, the medical instrument **100** is configured as a scope (e.g., endoscope, bronchoscope, ureteroscope, vascular scope, etc.). The scope can include, for example, a camera **136**, one or more light sources **138**, position sensors **140** (e.g., EM coils), and a working channel **142** as shown.

(146) The multi-modality construction of the elongated shaft **106** can include a plurality of modalities or properties. As used in this context, modality or property refers to a material or method of manufacture that can be varied in different regions along the length of the elongated shaft **106** to produce a variable bending stiffness profile. Examples will now be described with reference to the inner and outer layers **144**, **146** of the elongated shaft **106** of FIG. **16C**, although those of ordinary

skill in the art will appreciate that other modalities and properties can also be used in the construction of a medical instrument with a variable bending stiffness.

(147) For example, the mechanical properties of each section of the elongated shaft **106** can be modulated using the two-layer construction of FIG. **16C**. In this embodiment, the inner layer **144** comprises an endoskeleton **150**. As described below and shown in FIG. **17**, the endoskeleton **150** can comprise a laser cut hypotube made from a metal, such as nitinol or stainless steel. The material can vary depending on where in the elongated shaft **106** it is located. In some embodiments, the outer layer **146** can be a braided jacket. The braided jacket can include a braid **152** including the pull wire lumens and pull wires **108** and a thermoplastic material **154**, which can be melted into a composite structure. As discussed below, various properties of the braided jacket can be modulated in different regions of the elongated shaft to produce different bending stiffnesses. In some embodiments, the inner and outer layers **144**, **146** can be reversed (e.g., the endoskeleton **150** can be located in an outer layer).

(148) As shown in FIG. **16C**, the inner layer **144** and the outer layer **146** can be separated by a liner **148**. The liner **148** can be configured to keep the thermoplastic material **154** from the braided jacket from flowing into endoskeleton **150** during reflow operations during manufacturing and can allow the endoskeleton **150** to move freely under the braided jacket when in use.

(149) The endoskeleton **150** can be one modality that can be varied to develop a variable bending stiffness profile. For example, features of the endoskeleton **150** can be varied along its length to produce different bending stiffnesses in different regions of the elongated shaft **106**. In some embodiments, the endoskeleton **150** provides hoop strength and kink resistance for the elongated shaft **106**.

(150) The endoskeleton **150** can be formed, in some embodiments, of either nitinol or stainless steel, although other materials may also be used. In some embodiments, the endoskeleton **150** can be formed in part of nitinol and in another part of stainless steel. In some embodiments, the active bending section **114** of the elongated shaft **106** uses a nitinol flexure design, which allows it to go through tight bends. An example endoskeleton **150** including a nitinol flexure design **156** in the active bending section **114** is shown in FIG. **17**. The nitinol flexure design **156** can include various cut patterns designed to provide high axial stiffness to resist shrinkage during pull wire loads and low bending stiffness to reduce the force required to articulate the elongated shaft **106**. In some embodiments, use of a nitinol material may be advantageous in this section because of the high strain and tight bends this section experiences during use. The super elastic properties of nitinol allow it to spring back into a straight configuration even when highly articulated and resist fatigue even after many articulations.

(151) In some embodiments, the remainder of the endoskeleton **150** (e.g., the passive bending section **116**, the internal shaft section **118**, and the external shaft section **120**) can be formed of stainless steel, for example. In some sections (e.g., the passive bending section **116**, the internal shaft section **118**), the stainless steel can be laser cut with different pitches, wherein the lower the pitch, the lower the bending stiffness (and greater the flexibility). An example of this is shown, for example, by sections **158**, **160** of the endoskeleton **150** of FIG. **17**. In some embodiments, the laser cut sections may exhibit low axial stiffness, and a stiffer braided jacket in these sections can be used to increase bending stiffness. In the external shaft section **120**, the endoskeleton **150** can, in some embodiments, transition into a rigid hypotube, as illustrated by section **162** in FIG. **17**. This can provide the medical instrument **100** with maximum axial and bending stiffness in this area and prevent the elongated shaft **106** from buckling since this portion of the scope will often be unsupported outside of the sheath.

(152) In some embodiments, the elongated shaft **106** can include an endoskeleton **150** that advantageously provides an omni-directional or uniform bending stiffness circumferentially. The term omni-directional indicates that the elongated shaft **106** can have a bending stiffness that is the same in multiple directions. In some embodiments, the bending stiffness varies within 2-10 percent

about any direction of bending. The advantage of an omni-directional or uniform bending stiffness is that the performance of the elongated shaft **106** is indifferent to the orientation of the elongated shaft. In some embodiments, the elongated shaft **106** can have both two-way and one-way steering. (153) The endoskeleton **150** can provide various modalities that can be varied or modulated along the length of the elongated shaft to produce different bending stiffness in different sections. These modalities can include, for example, material, flexure design, laser cut, pitch, etc. These modalities can be varied along the length of the endoskeleton **150** to provide different bending stiffness in different regions or sections.

(154) The outer layer **146** can also include various modalities that can be varied or modulated along the length of the elongated shaft **106** to produce different bending stiffness in different sections. As noted previously, the outer layer **146** can comprise a braided jacket. In some embodiments, the braided jacket can be designed to house the pull wires **108** used for articulation of the bending section, provide mechanical structure and stability to the medical instrument, and seal the internals of the scope from the external environment.

(155) The braided jacket can include a jacket material **154** and braid **152**, with the pull wires **108** extending therethrough. In some embodiments, the braided jacket can provide mechanical structure and stability to the medical instrument **100**. The braided jacket can have a number of parameters that can be modulated to affect the properties of the elongated shaft **106**. For example, the jacket material **154** of the braided jacket can have a particular durometer or hardness. Different materials of different durometers can be used in different sections to produce different bending stiffnesses. In addition, the braids **152** of the braided jacket can be manufactured with a particular geometry or braid angle. The geometry and braid angle can be varied along the length of the elongated shaft **106** to produce different bending stiffnesses.

(156) In the illustrated embodiment, the mechanical properties of the braided jacket can be modulated by adjusting one or more of the following modalities, the durometer of the jacket material **154**, the braid **152** geometry, and the braid pic count (i.e., the braid angle). In general, lower durometer materials provide lower bending stiffness and higher durometer materials provide higher bending stiffness. Example materials that can be used for the braided jacket include polyether block amide (e.g., Pebax), available in a range of durometers from 25D-72D, and Nylon 12, which may be stiffer than Pebax. Other materials, such as plastics, can also be used. Regarding braid geometry, it is possible to adjust the shape and number of braids used in the braid layer. Those of skill in the art will appreciate that many different braid configurations possible. Finally, braid pic count (braid angle) can also be modulated. In some embodiments, the braided jacket may comprise additional modalities that can also be adjusted.

(157) The above described modalities are provided by way of example only. Other modalities may also be used, either in place of or in addition to the described modalities, to create the variable bending stiffness profile of the medical instrument **100**.

(158) FIG. **16D** schematically illustrates how various modalities can be modulated, according to one embodiment, in the multi-layer construction of the medical instrument **100** to create the bending stiffness profile **102** shown in FIG. **16B**. For reference, a side view of the medical instrument **100** is shown in FIG. **16D**, and the active bending section **114**, the passive bending section **116**, the internal shaft section **118**, and the external shaft section **120** are illustrated, as well as the first bending stiffness zone **122** of substantially constant bending stiffness, the first transition zone **124**, the second bending stiffness zone **126** of substantially constant bending stiffness, the second transition zone **128**, the third bending stiffness zone **130** of substantially constant bending stiffness, the third transition zone **132**, and the fourth bending stiffness zone **134** of substantially constant bending stiffness.

(159) In the illustrated embodiment, first and second layers of the medical instrument **100** are represented schematically. The first layer can be, for example, the inner layer **144** described above, and the second layer can be, for example, the outer layer **146**. As represented by the different cross-

hatching, a modality of the first layer changes along the length of the elongated shaft **106** between the distal end **112** and the proximal end **110**. In the illustrated example, the modality of the first layer comprises four distinct sections. Each of these sections can have a different property that makes a different contribution to the overall bending stiffness of the medical instrument **100**. As one example, the first layer can comprise the endoskeleton **150** and the four distinct sections of the first layer can represent the four sections **156, 158, 160, 162** shown in FIG. **17**. As shown in FIG. **16D**, transitions between the four sections of the first layer generally occur within transition zones **124, 128, 132**.

(160) In the illustrated embodiment, the second layer includes three different modalities. In some embodiments, the second layer can be a braided jacket, and the three modalities can represent the jacket material, braid geometry, and braid pic count as discussed above. In the illustrated example, the first modality includes four distinct sections. Each section can represent, for example, a material of different durometer or hardness. In the illustrated example, the second modality includes four distinct sections. Each section can represent, for example, a section with a different braid geometry. In the illustrated example, the third modality includes six distinct sections. Each section can represent, for example, a section of different pic count. In general, transitions between sections occur within transition zones **124, 128, 132**, although as shown, for example, in the second and third modalities, this need not always be the case.

(161) Together the arrangement of different sections of the various modalities of the first and second layer define the bending stiffness of the elongated shaft **106** of the medical instrument **100**. In some embodiments, to achieve bending stiffness zones having a substantially constant bending stiffness (e.g., bending stiffness zones **122, 126, 130, 134**), the transitions between the different sections of the various modalities of the first and second layer generally do not occur or are limited within the bending stiffness zones **122, 126, 130, 134**. Because the construction of the elongated shaft **106** generally remains constant in these bending stiffness zones, the bending over the length of these zones can be substantially constant. In some embodiments, no transition between different sections of the various modalities of the first and second layer occurs within a bending stiffness zone of substantially constant bending stiffness. In some embodiments, at most one, two, or three transitions between different sections of the various modalities of the first and second layer occur within a bending stiffness zone of substantially constant bending stiffness.

(162) In some embodiments, to achieve transition zones having a length over which the bending stiffness of the zone gradually transitions from a first bending stiffness to a second bending stiffness (e.g., transition zones **124, 128, 132**), transitions between the different sections of the various modalities of the first and second layer generally occur within the transition zone. For example, in some embodiments, at least one, two, three, four, five, six, or more transitions between the different sections of the various modalities of the first and second layer generally occur within a transition zone. Additionally, to create a gradual (e.g., sloped or ramped) transition within a transition zone, the transitions between the different sections of the various modalities of the first and second layer can be generally staggered. That is, within a transition zone, transitions between the different sections of the various modalities of the first and second layer generally occur within the zone generally do not occur at the same point along the length of the elongated shaft **106**. This can have a smoothing effect on the overall bending stiffness profile **102** of the medical instrument **100** within the transition zones **124, 128, 132**. In some embodiments, at least two, three, four, five, six, or more transitions between the different sections of the various modalities of the first and second layer are staggered within each transition zone.

(163) FIG. **16D** illustrates that in some embodiments, the medical instrument **100** can comprise a multi-layer construction. Some layers may comprise a single modality (e.g., the first layer) and some layers may comprise a plurality of modalities (e.g., the second layer). In some embodiments, the medical instrument may comprise only a single layer. In some embodiments, the medical instrument may comprise more than two layers. Further, the illustrated modalities and transitions

between sections illustrated in FIG. 16D are provided by way of example only. Those of skill in the art will readily appreciate that other arrangements are possible to produce various bending stiffness profiles as desired.

(164) FIGS. 18A-18D relate to an embodiment of another medical instrument **200** having an example variable bending stiffness profile **202**. In the illustrated embodiment, the medical instrument **200** is configured as a sheath. As will be described below, in some embodiments, the medical instrument **200** (sheath) can be used with the medical instrument **100** (scope) to form a compound structure, and relative movement of the medical instrument **200** (sheath) and the medical instrument **100** (scope) can be used to modulate the bending stiffness profile of the compound structure. FIG. 18A is a side view of the medical instrument **200**. FIG. 18B illustrates the variable bending stiffness profile **202** of the medical instrument **200**. FIG. 18C illustrates a cross-sectional view of the medical instrument **200**, showing an example multi-modality construction thereof. FIG. 18D schematically illustrates an example of how the individual modalities of the multi-modal construction can be varied to achieve the variable bending stiffness profile **202** of the medical instrument **200**.

(165) In many respects, the medical instrument **200** is similar to the medical instrument **100**. As illustrated in FIG. 18A, the medical instrument **200** includes an instrument base **204** and an elongated shaft **206**. The elongated shaft **206** can extend between a proximal end **210** and a distal end **212**. The proximal end **210** can extend from and be attached to the instrument base **204**. The distal end **212** can be the leading end of the elongated shaft **206** of the medical instrument **200**. The elongated shaft **206** includes a length L measured between the proximal end **210** and the distal end **212**. In some embodiments, the length L of the medical instrument **200** (sheath) is less than the length L of the medical instrument **100** (scope), while in other embodiments, the length L of the medical instrument **200** (sheath) is equal to or greater than the medical instrument **100** (scope). In some embodiments, the length L of the medical instrument **200** (sheath) is about 50-300 mm less than the length L of the medical instrument **100** (scope). In some embodiments, the length L of the medical instrument **200** (sheath) is about 100 mm, about 150 mm, about 200 mm, about 250 mm, or about 300 mm less than the length L of the medical instrument **100** (scope). In some embodiments, the length L of the medical instrument **200** (sheath) is about 60%, about 70%, about 80%, or about 90% the length L of the medical instrument **100** (scope). In some embodiments, for example, the length L of the medical instrument **200** may be between about 600 and about 750 mm. In one example, the length L is about 680 mm.

(166) In the illustrated embodiment, the elongated shaft **206** includes three different sections: a bending section **214** (also referred to an active bending section), a passive bending section **216**, and an internal shaft section **218** (also referred to as a tracking section). The sections can be arranged serially along the length L of the elongated shaft **206** between the distal end **212** and the proximal end **210**. Although three sections are illustrated, in some embodiments, a medical instrument can include more or less sections. For example, a medical instrument can include one, two, three, four, five, six, seven, or more sections.

(167) Similar to the medical instrument **100**, each section of the medical instrument **200** can be configured to provide a different function to facilitate the medical procedure and/or improve the drivability and control of the medical instrument **200**. For example, in some embodiments, more distally located sections are more flexible to facilitate maneuverability of the elongated shaft **206**, and more proximally located sections are stiffer to provide pushability of the elongated shaft **206**, while preventing or reducing the likelihood of buckling. In some embodiments, the active bending section **214** provides steerability to the distal end of the medical instrument **200**. In some embodiments, the passive bending section **216** provides enough flexibility to track up in to the upper lobes of the lung, for example, in the case of the bronchoscopy. In some embodiments, the internal shaft section **218** is flexible enough to be inserted through an introducer and tube bends, but pushable/rigid enough to provide support for the elongated shaft **206**. In some embodiments, to

achieve these different functions, each section comprises different bending stiffness characteristics or properties.

(168) In some embodiments, the active bending section **214** has a length between about 50 mm and 80 mm, or approximately 65 mm. In some embodiments, the passive bending section **216** has a length between about 45 mm and 75 mm, or approximately 60 mm. In some embodiments, the internal shaft section **218** can have a length between about 410-500 mm, or approximately 450 for the internal zone. Advantageously, the different lengths of the three different zones are designed to desirably align the elongated shaft **206** with portions of the underlying scope, thereby creating a compound structure with desired variable stiffnesses along different lengths. In some embodiments, the lengths of one, more, or all of these sections can be different than the listed values.

(169) FIG. **18B** illustrates a representative bending stiffness profile **202** of the elongated shaft **206** of the medical instrument **200**. In the illustrated embodiment and as seen in the example bending stiffness profile **202**, the elongated shaft **206** of the medical instrument **200** includes three bending stiffness zones of substantially constant bending stiffness and two transition zones, with each transition zone positioned between a pair of adjacent bending stiffness zones. In the illustrated example, the medical instrument **200** includes (arranged distally to proximally) a first bending stiffness zone **222** of substantially constant bending stiffness, a first transition zone **224**, a second bending stiffness zone **226** of substantially constant bending stiffness, a second transition zone **228**, and a third bending stiffness zone **230** of substantially constant bending stiffness.

(170) Although three bending stiffness zones and two transition zones are shown and described, other embodiments may include other numbers of bending stiffness zones and transmissions. For example, a medical instrument could include two, three, four, five, six, or more bending stiffness zones of substantially constant bending stiffness, separated by one, two, three, four, five, or more transition zones.

(171) In the illustrated embodiment, the first bending stiffness zone **222** has the lowest substantially constant bending stiffness, the second bending stiffness zone **226** has a substantially constant bending stiffness that is higher than that of the first bending stiffness zone **222**, and the third bending stiffness zone **230** has a substantially constant bending stiffness that is higher than that of the second bending stiffness zone **226**. Thus, the bending stiffness of the medical instrument **200** generally increases from the distal end **212** to the proximal end **210**. This need not be the case in all embodiments. For example, the bending stiffness could increase, decrease, and then increase again or generally decrease.

(172) Further, in the illustrated embodiment, the transition zones **224**, **228** include generally or substantially constant slopes. It will be appreciated that the slope of each transition zone can vary. For example, the slope of the first transition zone **224** can be less than the slope of the second transition zone **228**. Again, this need not be the case in all embodiments.

(173) For reference, FIG. **18B** also illustrates how, in one example, the bending stiffness profile **202** aligns with the bending section **214**, the passive bending section **216**, and the internal shaft section **218**. In the illustrated embodiment, each section includes one bending stiffness zone of substantially constant bending stiffness. For example, the first bending stiffness zone **222** is positioned in the bending section **214**, the second bending stiffness zone **226** is positioned in the passive bending section **216**, and the third bending stiffness zone **230** is positioned in the internal shaft section **218**. The elongated shaft **206** can be configured such that the substantially constant bending stiffness of the bending stiffness zone positioned within the section facilitates or enables the section to perform its specified function. The transition zones **224**, **228** provide lengths over which the bending stiffness of the elongated shaft can gradually vary from a lower value to a higher value (or vice versa). As before, having regions where the bending stiffness varies gradually (as opposed to abruptly) can help to avoid stress risers and failure points within the elongated shaft **206**.

(174) In the illustrated embodiment, the transition zones **224**, **228** are positioned to overlap

adjacent sections. For example, the first transition zone **224** is positioned between and partially overlaps the bending section **214** and the passive bending section **216**. The second transition zone **228** is positioned between and partially overlaps the passive bending section **216** and the internal shaft section **218**. Other arrangements and positions for the transition zones are also possible. For example, a transition zone could be positioned entirely within one section, for example, at a distal end of the section, at a proximal end of the section, or along the length of a section between the proximal and distal ends.

(175) FIG. **18C** is a cross-sectional view of the elongated shaft **206** of medical instrument **200** including a multi-modal construction configured to provide a variable bending stiffness profile **202** as shown, for example, in FIG. **18C**. The multi-modal construction of the medical instrument **200** can comprise a plurality of layers. For example, as illustrated, the medical instrument **200** comprises an inner layer **244** and an outer layer **246**. The inner and outer layers **244**, **246** can surround a working or inner channel **242** as shown. The inner channel **242** can be configured in size and shape to receive a scope (e.g., the medical instrument **100**) therein. In some embodiments, the scope can telescope through the inner channel **242**. A liner **248** can be provided around the inner channel **242**. The liner **248** can be configured to allow the first medical instrument **100** to move freely through the inner channel **242** during use.

(176) In the illustrated embodiment, both the inner layer **244** and the outer layer **246** can comprise a braided jacket. The braided jacket can include a braid made from a thermoplastic material. In some embodiments one of the inner layer **244** and the outer layer **246** includes pull wires **208**. As discussed above, various properties of the braided jacket can be modulated in different regions of the elongated shaft to produce different bending stiffnesses. For example, in each layer **244**, **246**, braid material (durometer), braid geometry, and braid pic count can be modulated to produce different bending stiffnesses. Other modalities may also be used, either in place of or in addition to the described modalities, to create the variable bending stiffness profile of the medical instrument **200**.

(177) FIG. **18D** schematically illustrates how the various modalities can be modulated, according to one embodiment, in the multi-layer construction of the medical instrument **200** to create the bending stiffness profile **202** shown in FIG. **18B**. For reference, a side view of the medical instrument **200** is shown in FIG. **18D**, and the active bending section **214**, the passive bending section **216**, and the internal shaft section **218** are illustrated, as well as the first bending stiffness zone **222** of substantially constant bending stiffness, the first transition zone **224**, the second bending stiffness zone **226** of substantially constant bending stiffness, the second transition zone **228**, and the third bending stiffness zone **230** of substantially constant bending stiffness.

(178) In the illustrated embodiment, first and second layers of the medical instrument **200** are represented schematically. The first layer can represent the inner layer **244** and the second layer can represent the outer layer **246**, which, as described above, can both be braided jackets. As illustrated in FIG. **18D**, both the first layer and the second layer can include at least two different modalities. In the illustrated example, the first modality of the first layer includes four distinct sections, and the second modality of the first layer includes four distinct sections. In the illustrated example, the first modality of the second layer includes four distinct sections, and the second modality of the second layer includes six distinct sections. As with the medical instrument **100**, in general, transitions between sections do not occur or are limited in bending zones **222**, **226**, **230** of substantially constant bending stiffness, and transitions between sections occur within transition zones **224**, **228**, although this need not always be the case. Other numbers of layers, having different numbers of modalities and different numbers of sections can be used.

B. Modulating Bending Stiffness Profiles

(179) As mentioned briefly above, in some embodiments, a compound structure (sometimes referred to as a medical instrument) can comprise a scope (e.g., the medical instrument **100**) positioned within an inner channel of a sheath (e.g., the medical instrument **200**). As noted above,

the term “compound structure” can refer to and encompass all parts of the scope and the sheath whenever the scope is positioned within the inner channel of the sheath, including parts of the scope that may extend distally from the sheath, or parts of the sheath that may extend distally from the scope. One or both of the scope and sheath can comprise a variable bending stiffness profile (see, e.g., FIGS. **16B** and **18B**) as described above. The compound structure can comprise a variable bending stiffness profile that is a combination of the individual variable bending stiffness profiles of the scope and sheath. Further, the variable bending stiffness profile of the compound structure can be modulated, changed, or varied by adjusting the relative positions of the scope and the sheath. In some embodiments, modulation of the variable bending stiffness profile of the compound structure can occur during a medical procedure (e.g., intraoperatively), either manually or robotically, so as to facilitate the medical procedure.

(180) FIG. **19A** illustrates a medical instrument or compound structure **300**. The compound structure **300** includes a scope **100**, for example, the medical instrument **100** of FIGS. **16A-16D**, positioned within the inner channel of a sheath **200**, for example, the medical instrument **200** of FIGS. **18A-18D**. Specifically, the elongated shaft **106** of the scope **100** can extend through the inner channel **242** of the elongated shaft **206** of the sheath **200**. In some embodiments, the scope **100** and the sheath **200** are coaxial. In the illustrated embodiment of FIG. **19A**, the scope **100** and the sheath **200** are positioned relative to each other such that the distal end **112** of the scope **100** is aligned with the distal end **212** of the sheath **200**.

(181) While the scope **100** and the sheath **200** may be coaxial, they can be independently controlled relative to one another. For example, the scope **100** includes an instrument base **104** that can be attached to an instrument drive mechanism on a first robotic arm. The first robotic arm can move the instrument base **104** to insert or retract the scope **100** in the directions **170**. Similarly, the sheath **200** includes an instrument base **204** that can be attached to an instrument drive mechanism on a second robotic arm. The second robotic arm can move the instrument base **204** to insert or retract the sheath **200** in the directions **270**. Movement of the scope in the directions **170** can be independent of movement of the sheath **200** in the directions **270**. In some embodiments, the scope **100** and the sheath **200** are not be coextensive in length, such that one (typically the sheath **200**) will be shorter than the other (typically the scope **100**).

(182) FIG. **19B** illustrates example bending stiffness profiles **102**, **202**, **302** of the scope **100**, the sheath **200**, and the compound structure **300**, respectively. The scope **100** may comprise a variable bending stiffness profile **102** as described above (see, e.g., FIG. **16B**). In the illustrated example, the bending stiffness profile **102** of the scope **100** includes four bending stiffness zones of substantially constant bending stiffness and three transition zones. The sheath **200** may also comprise a variable bending stiffness profile **202** as described above (see, e.g., FIG. **18B**). In the illustrated example, the bending stiffness profile **202** of the sheath **200** includes three bending stiffness zones of substantially constant bending stiffness and two transition zones.

(183) As shown in FIG. **19B**, the compound structure **300** includes a bending stiffness profile **302** that is based on the bending stiffness profiles **102**, **202** of the scope **100** and the sheath **200**. In the illustrated embodiment, the bending stiffness profile **302** includes five bending stiffness zones of substantially constant bending stiffness and four transition zones. Thus, in some embodiments, the bending stiffness profile **302** of the compound structure **300** comprises more bending stiffness zones of substantially constant bending stiffness and/or more transition zones than one or both the bending stiffness profiles **102**, **202** of the scope **100** and the sheath **200**. In some embodiments, the bending stiffness profile **302** of the compound structure **300** comprises the same number of bending stiffness zones of substantially constant bending stiffness and/or more transition zones as one or both the bending stiffness profiles **102**, **202** of the scope **100** and the sheath **200**.

(184) For example, if the scope **100** comprises n bending stiffness zones, the compound structure **300** can, in some embodiments, comprise any number of bending stiffness zones between $n+1$ and $2n$ (inclusive) or more. As another example, if the sheath **200** comprises m bending stiffness zones,

the compound structure **300** can, in some embodiments, comprise any number of bending stiffness zones between $m+1$ and $2m$ (inclusive) or more. As an additional example, if the scope comprises n bending stiffness zones and the sheath **200** comprises m bending stiffness zones, the compound structure **300** can, in some embodiments, comprise any number of bending stiffness zones between $n+m+1$ and $2(n+m)$ (inclusive) or more.

(185) Further, the bending stiffness profile **302** of the compound structure **300** can be modulated, changed, or varied, by adjusting the relative positions of the scope **100** and sheath **200**. FIG. **19C** illustrates an example, where the position of the scope **100** has been adjusted relative to the sheath **200**. Specifically, in the illustrated example, the distal end **112** of the scope **100** has been positioned beyond (more distal of) the distal end **212** of the sheath **200**. This can be accomplished, for example, by advancing the scope **100**, retracting the sheath **200**, or performing a combination thereof.

(186) FIG. **19D** illustrates the bending stiffness profiles **102**, **202**, **302** of the scope **100**, the sheath **200**, and the compound structure **300**, respectively, with the scope **100** and the sheath **200** positioned as shown in FIG. **19C**. In the illustrated example, the bending stiffness profile **302** of the compound structure **300** comprises six bending stiffness zones of substantially constant bending stiffness separated by transition zones. Further, it should be appreciated that the bending stiffness profile **302** of FIG. **19D** has been modulated or changed relative to the bending stiffness profile **302** of FIG. **19B**. This is because, with the scope **100** and the sheath **200** in different positions, the bending stiffness profiles **102**, **202** line up in a different way resulting in a different bending stiffness profile **302**.

(187) Thus, the compound structure **300** can advantageously modulate or change its bending stiffness profile **302** based on a plurality of relative positions of the scope **100** and the sheath **200**, including relative positions where the distal end **112** of the scope **100** is positioned beyond the distal end **212** of the sheath **200**, relative positions where the distal end **212** of the sheath **200** is positioned beyond the distal end **112** of the scope **100**, and the relative position where the distal end **112** of the scope **100** is aligned with the distal end **212** of the sheath **200**.

(188) In some embodiments, moving the scope **100** relative to the sheath **200** (or vice versa) can result in a bending stiffness profile **302** of the compound structure that has more, fewer, or the same number of bending stiffness zones of substantially constant bending stiffness as prior to the movement.

(189) In some embodiments, modulation of the scope **100** by the sheath **200** can occur only along a portion of the scope **100** that is positioned within the sheath **200**, with the remainder of the scope **100** (e.g., the portion extending distally of the sheath **200**) having its original bending stiffness profile, as shown in FIG. **19D**.

(190) During a medical procedure, a physician and/or a robotically enabled medical system can adjust the relative position of the scope **100** and the sheath **200** to modulate the bending stiffness profile **302** of the compound structure. This can be done to produce a bending stiffness profile **302** that is more suited for a particular maneuver to be performed. In some embodiments, this is performed manually. In some embodiments, this is performed automatically (e.g., as controlled by a processor or navigation system). In some embodiments, this is performed intraoperatively. This may advantageously allow use of different bending stiffness profiles during different instances of a procedure without needing to change medical instruments.

(191) FIGS. **20A-20C** present experimental data related to a first embodiment of a compound structure (scope and sheath) configured according to the principles described herein. In this example, both the scope and the sheath comprise a variable bending stiffness profile. FIG. **20A** illustrates the bending stiffness profiles of the scope and sheath, as well as a combined bending stiffness profile of the compound structure when distal ends of the scope and sheath are aligned. FIG. **20B** illustrates the bending stiffness profiles of the scope and sheath, as well as a combined bending stiffness profile of the compound structure when the distal end of the scope is extended 60

mm beyond the distal end of the sheath. FIG. 20C illustrates the bending stiffness profiles of the scope and sheath, as well as a combined bending stiffness profile of the compound structure when the distal end of the scope is extended 150 mm beyond the distal end of the sheath. Comparing the combined bending stiffness profiles of the compound structure shown in FIGS. 20A-20C it can be seen that the combined bending stiffness profile can be modulated by adjusting the relative position of the scope and the sheath.

(192) FIGS. 21A and 21B present experimental data related to a second embodiment of a compound structure (scope and sheath) configured according to the principles described herein. In this example, both the scope and the sheath comprise a variable bending stiffness profile. FIG. 21A illustrates variable stiffness profiles for the compound structure at different extensions of the scope relative to the sheath. As shown, all five variable stiffness profiles of the compound structure are different. FIG. 21B illustrates the variable stiffness profile of the compound structure as a function of extension of the scope relative to the sheath. Again, it is shown that adjusting the position of the scope relative to the sheath modulate the bending stiffness profile of the compound structure.

(193) FIG. 23 is a flowchart illustrating an example method 400 for navigating a medical instrument within a patient lumen. In some embodiments, the medical instrument may comprise a scope and sheath, such as the compound structure 300 described above. The method 400 can be implemented in a robotic medical system. In some embodiments, the method 400 can be implemented manually.

(194) The method 400 begins at block 402. At block 402, the medical instrument is inserted into a patient lumen. The medical instrument may comprise a scope and sheath. The scope may comprise a plurality of bending stiffness zones arranged along a length of the scope. In some embodiment, each of the bending stiffness zones has a bending stiffness that is substantially uniform. The sheath may comprise a plurality of bending stiffness zones arranged along a length of the sheath and an inner channel. The scope can be positioned within the inner channel of the sheath. The patient lumen can comprise a bronchial airway, although the method 400 may be implemented for use with other patient lumens as well.

(195) Next, at block 404, the bending stiffness profile of the medical instrument can be modulated by adjusting the position of at least one of the scope and the sheath relative to the other of the scope and the sheath. In some embodiments, modulating the bending stiffness profile of the medical instrument comprises at least one of advancing the scope and retracting the sheath such that the scope extends distally from the sheath. In some embodiments, modulating the bending stiffness profile of the medical instrument comprises at least one of advancing the sheath and retracting the scope such that a distal end of the scope is positioned within the inner channel of the sheath. In some embodiments, a distal end of the sheath can be positioned distally beyond the distal end of the scope. In some embodiments, the distal end of the scope is aligned with a distal end of the sheath.

(196) In some embodiments, the scope comprises at least four bending stiffness zones, each having a substantially constant bending stiffness along a length of the bending stiffness zone. In some embodiments, the scope comprises at least three transition zones, each of the at least three transition zones positioned between a pair of the at least four bending stiffness zones. In some embodiments, the scope comprises at least three bending stiffness zones, each having a substantially constant bending stiffness along a length of the bending stiffness zone. In some embodiments, the scope comprises at least two transition zones, each of the at least two transition zones positioned between a pair of the at least three bending stiffness zones. In some embodiments, a first bending stiffness zone has a bending stiffness that is less than a bending stiffness of the other bending stiffness zones. In some embodiments, a third zone of bending stiffness extends through an introducer. In some embodiments, a fourth zone of bending stiffness does not extend through the patient lumen, nor through the introducer.

3. Implementing Systems and Terminology

(197) Implementations disclosed herein provide systems, methods and apparatus for medical instruments with variable bending stiffness profiles.

(198) It should be noted that the terms “couple,” “coupling,” “coupled” or other variations of the word couple as used herein may indicate either an indirect connection or a direct connection. For example, if a first component is “coupled” to a second component, the first component may be either indirectly connected to the second component via another component or directly connected to the second component.

(199) The position estimation and robotic motion actuation functions described herein may be stored as one or more instructions on a processor-readable or computer-readable medium. The term “computer-readable medium” refers to any available medium that can be accessed by a computer or processor. By way of example, and not limitation, such a medium may comprise random access memory (RAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), flash memory, compact disc read-only memory (CD-ROM) or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer. It should be noted that a computer-readable medium may be tangible and non-transitory. As used herein, the term “code” may refer to software, instructions, code or data that is/are executable by a computing device or processor.

(200) The methods disclosed herein comprise one or more steps or actions for achieving the described method. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is required for proper operation of the method that is being described, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims.

(201) As used herein, the term “plurality” denotes two or more. For example, a plurality of components indicates two or more components. The term “determining” encompasses a wide variety of actions and, therefore, “determining” can include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” can include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” can include resolving, selecting, choosing, establishing and the like.

(202) The phrase “based on” does not mean “based only on,” unless expressly specified otherwise. In other words, the phrase “based on” describes both “based only on” and “based at least on.”

(203) As used herein, the term “approximately” or “about” refers to a range of measurements of a length, thickness, a quantity, time period, or other measurable value. Such range of measurements encompasses variations of $\pm 10\%$ or less, preferably $\pm 5\%$ or less, more preferably $\pm 1\%$ or less, and still more preferably $\pm 0.1\%$ or less, of and from the specified value, in so far as such variations are appropriate in order to function in the disclosed devices, systems, and techniques.

(204) The previous description of the disclosed implementations is provided to enable any person skilled in the art to make or use the present invention. Various modifications to these implementations will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the scope of the invention. For example, it will be appreciated that one of ordinary skill in the art will be able to employ a number corresponding alternative and equivalent structural details, such as equivalent ways of fastening, mounting, coupling, or engaging tool components, equivalent mechanisms for producing particular actuation motions, and equivalent mechanisms for delivering electrical energy. Thus, the present invention is not intended to be limited to the implementations shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. A compound medical instrument comprising: an elongated shaft extending between a distal end and a proximal end, the elongated shaft comprising: at least three bending stiffness zones arranged along a length of the elongated shaft, each bending stiffness zone comprising a bending stiffness that extends over a length of the bending stiffness zone, wherein the bending stiffness of each of the at least three bending stiffness zones is greater or less than the bending stiffness of an adjacent bending stiffness zone, and wherein the bending stiffness of each of the at least three bending stiffness zones is substantially uniform along its length; and at least two transition zones, wherein each transition zone comprises a variable bending stiffness extending over a length of the transition zone between each pair of the at least three bending stiffness zones, wherein the variable bending stiffness of each transition zone varies from a first bending stiffness on a first side of the transition zone to a second bending stiffness on a second side of the transition zone; a sheath having a sheath length extending between a sheath distal end and a sheath proximal end, the sheath comprising an inner channel in which the elongated shaft is disposed, the sheath further comprising bending stiffness zones arranged along the length of the sheath; and a robotic system configured to control movement of the elongated shaft and the sheath independent of each other, the elongated shaft and the sheath forming a compound structure, the robotic system comprising a processor including a computer readable medium having instructions stored thereon that when executed cause the processor to: determine an intraoperative maneuver of a medical procedure; and in response to determining the intraoperative maneuver, adjust a relative position of the elongated shaft and the sheath to modulate a bending stiffness profile of the compound structure based on the maneuver.
2. The compound medical instrument of claim 1, wherein, for each transition zone, the variable bending stiffness varies from the first bending stiffness on the first side of the transition zone to the second bending stiffness on the second side of the transition zone with a substantially linear slope.
3. The compound medical instrument of claim 1, wherein, of the at least three bending stiffness zones arranged along the length of the elongated shaft, the bending stiffness zone closest to the distal end has a bending stiffness that is less than a bending stiffness of each of the two other bending stiffness zones.
4. The compound medical instrument of claim 1, wherein, of the at least three bending stiffness zones arranged along the length of the elongated shaft, the bending stiffness zone closest to the proximal end has a bending stiffness that is greater than a bending stiffness of each of the two other bending stiffness zones.
5. The compound medical instrument of claim 1, wherein the bending stiffness of each of the at least three bending stiffness zones increases from the distal end to the proximal end of the elongated shaft.
6. The compound medical instrument of claim 1, wherein each of the at least three bending stiffness zones has a length of at least 50 mm.
7. The compound medical instrument of claim 1, wherein each of the transition zones has a length of at least 10 mm.
8. The compound medical instrument of claim 1, wherein the elongated shaft comprises an inner layer and an outer layer.
9. The compound medical instrument of claim 8, wherein the outer layer comprises a braided jacket and the inner layer comprises an endoskeleton.
10. The compound medical instrument of claim 9, wherein the braided jacket comprises one or more pull wires extending therethrough.
11. The compound medical instrument of claim 9, wherein a bending stiffness of the braided jacket is modulated by at least one of the following: jacket material durometer, braid geometry, and braid angle.
12. The compound medical instrument of claim 9, wherein the endoskeleton comprises a first portion formed of nitinol and a second portion formed of stainless steel.

13. The compound medical instrument of claim 12, wherein the second portion of the endoskeleton comprises a first section comprising coils having at least a first pitch and a second section comprising coils having at least a second pitch.
14. The compound medical instrument of claim 13, wherein the second portion of the endoskeleton further comprises a third section formed as a rigid hypotube.
15. The compound medical instrument of claim 1, wherein the computer readable media further includes instructions stored thereon, that when executed, causes the processor to calculate an extension of the elongated shaft relative to the sheath that produces the bending stiffness profile of the compound structure.
16. The medical instrument of claim 1, wherein the elongated shaft includes one or more pull wires, and the processor is further configured to actuate the one more pull wires to change a shape or pose of the compound structure after modulating the bending stiffness profile.
17. A medical instrument comprising: an elongated shaft extending between a distal end and a proximal end, the elongated shaft comprising a plurality of bending stiffness zones arranged along a length of the elongated shaft and at least two transition zones, wherein each transition zone comprises a variable bending stiffness extending over a length of the transition zone between adjacent bending stiffness zones; a sheath having a sheath length extending between a sheath distal end and a sheath proximal end, the sheath comprising an inner channel in which the elongated shaft is disposed, the sheath further comprising bending stiffness zones arranged along the length of the sheath; and a robotic system configured to control movement of the elongated shaft and the sheath independent of each other, the elongated shaft and sheath forming a compound structure, the robotic system comprising a processor including a computer readable medium having instructions stored thereon, that when executed, causes the processor to: determine an intraoperative maneuver of a medical procedure; and in response to determining the intraoperative maneuver, adjust a relative position of the elongated shaft and the sheath to modulate a bending stiffness profile of the compound structure based on the maneuver.
18. The medical instrument of claim 17, wherein the elongated shaft includes one or more pull wires, and the processor is further configured to actuate the one more pull wires to change a shape or pose of the compound structure after modulating the bending stiffness profile.
19. The medical instrument of claim 17, wherein the robotic system further comprises: a first robotic arm coupled to the compound structure and configured to translate the sheath to one or more first positions along a longitudinal axis of the compound structure; and a second robotic arm coupled to the compound structure and configured to translate the elongated shaft to one or more second positions along the longitudinal axis of the compound structure independent of the movement of the sheath, wherein the first and second arms are coordinated to adjust the relative position of the elongated shaft and the sheath to modulate the bending stiffness profile of the compound structure.
20. The medical instrument of claim 17, wherein adjusting the relative position of the elongated shaft and the sheath includes changing an axial position of both the sheath and the elongated shaft along a longitudinal axis of the compound structure.
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